

STABILITY FOR BOUNDARY ACTIONS OF COCOMPACT LATTICES IN EUCLIDEAN BUILDINGS

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ABSTRACT. When X is a locally compact Euclidean building, the isometry group of X acts by homeomorphisms on the space of k -simplices in the visual boundary of X . We consider perturbations of these actions for discrete groups of isometries acting with compact quotient on X , showing that all small enough perturbations are semi-conjugate to the original action. This proves in particular that, when Q is any parabolic subgroup in a semisimple p -adic Lie group G , the induced action of a cocompact lattice in G has a topologically stable action on G/Q .

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1. INTRODUCTION

Zimmer's program [Zim87] seeks to understand and classify actions of large groups, particularly higher-rank lattices, on manifolds. While substantial progress has been achieved for lattices in real Lie groups (for examples [Sul85, Gly93, Kan96, KS97, FS00, FM05, BFH16, BFH20, Pec20, BFH21, DSVX22, KKL19, CINS23, Pec24, BRW24]), much less is known in the p -adic setting. One of the fundamental obstacles is the absence of a satisfactory theory of smooth dynamics for totally disconnected groups. In the real case, many rigidity arguments rely on differentiability and on the structure theory of Lie groups, whose foundations originate in the solution of Hilbert's Fifth Problem. The p -adic setting naturally leads to analogous questions concerning actions of totally disconnected locally compact groups on manifolds. Closely related is the Hilbert–Smith Conjecture, which predicts that if a locally compact group acts faithfully on a connected manifold, then it must be a Lie group; equivalently, the p -adic integers $\mathbb{Z}_p$ cannot act faithfully on a manifold. Despite significant efforts, there has been relatively little progress toward a p -adic analogue of Zimmer's program for actions on manifolds.

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A natural approach is to replace manifolds by geometric spaces that play the role of homogeneous manifolds in the p -adic world. For semisimple p -adic groups (a semisimple Lie group over a non-Archimedean local field), flag varieties and their associated boundaries provide such a class of spaces. These spaces carry rich geometric and combinatorial structures, arise naturally from the theory of algebraic groups and buildings, and serve as p -adic counterparts of flag manifolds and homogeneous spaces in the real setting. In this paper, we study actions of lattices and other large subgroups on p -adic flag spaces and investigate the rigidity phenomena that arise. Our goal is to understand to what extent the strong rigidity properties predicted by Zimmer’s program persist in this non-Archimedean setting and to develop new techniques adapted to the geometry of p -adic groups and their boundaries.

We work with p -adic flag spaces using their connection to *spherical* and *Euclidean buildings*, and heavily exploit the $\text{CAT}(k)$ geometry of these buildings. This links our result to the general theory of “actions at infinity” for groups of isometries of a “nonpositively curved” metric space X —for example, the action of a Gromov-hyperbolic group on its Gromov boundary, or a $\text{CAT}(0)$ group on its visual boundary. Since such boundary actions often encode detailed features of the original action on X , it is natural to ask what features of some boundary action $\Gamma \rightarrow \text{Homeo}(\partial X)$ are perturbed in the space $\text{Hom}(\Gamma, \text{Homeo}(\partial X))$. Questions of this form have a long history, and some of their answers form an important part of rigidity theory, dating back to [Sul85, Ghy93, Kan96, KS97]. They are further motivated by the crucial role that induced boundary maps play in the celebrated rigidity results of Mostow [Mos68] and Margulis [Mar69].

For the statement of our main result, let X be a *locally compact Euclidean building*, i.e. a $\text{CAT}(0)$ space locally modeled on a Euclidean Coxeter complex (a_{mod}, W) ; see [KL97] for background. The visual boundary of this space has the structure of a *spherical building*, meaning it decomposes as a union of simplices, each canonically identified with a *model simplex* σ_{mod} . Each face $\tau_{\text{mod}} \subseteq \sigma_{\text{mod}}$ therefore defines a *space of simplices* in $\partial_{\infty} X$, giving a finite family of different bordifications of X . The corresponding boundaries are called the *flag boundaries* for the Euclidean building X . Each flag boundary M is a compact space homeomorphic to a Cantor set, and any subgroup $\Gamma < \text{Isom}(X)$ acts by homeomorphisms on M . We consider perturbations of this Γ -action, and prove the following theorem:

Theorem 1.1. *Let X be a locally compact Euclidean building, let Γ be a group acting properly and cocompactly by isometries on X , and let $\rho_0 : \Gamma \rightarrow \text{Homeo}(M)$ denote the induced action of Γ on the space M of τ_{mod} -flags. Then ρ_0 is topologically stable, i.e. there is a neighborhood U of ρ_0 in $\text{Hom}(\Gamma, \text{Homeo}(M))$ consisting of actions semi-conjugate to ρ_0 .*

Here we say that two actions $\rho, \rho' : \Gamma \rightarrow \text{Homeo}(M)$ are *semi-conjugate* if there exists a (ρ, ρ') -equivariant surjection $M \rightarrow M$. Semi-conjugacy is the appropriate notion of equivalence when considering C^0 perturbations of an action $\Gamma \rightarrow \text{Homeo}(M)$; thus, the theorem above can also be viewed as a local rigidity result for the “standard” Γ -action ρ_0 on M .

Remark 1.2. Semi-conjugacy is the optimal conclusion in the setting of Theorem 1.1: in Section 7 we give an example showing that there can exist arbitrarily small perturbations of ρ_0 that are *not* conjugate to ρ . On the other hand, if one asks for the perturbed actions to be bi-Lipschitz, then, by the same argument as in [CINS23, Section 4.4], the semi-conjugacy can be upgraded to a conjugacy.

When G is a semisimple Lie group over a non-Archimedean local field, then G acts properly discontinuously by isometries on a locally compact Euclidean building X as above, and the different flag boundaries of X are canonically identified with the quotient spaces G/Q for the various parabolic subgroups $Q < G$. So Theorem 1.1 implies:

Corollary 1.3. *Let G be a semisimple Lie group over a non-Archimedean local field, let $Q < G$ be a parabolic subgroup, and let Γ be a lattice in G . The induced action $\rho_0 : \Gamma \rightarrow \text{Homeo}(G/Q)$ is topologically stable.*

Note that a lattice in a p -adic Lie group is necessarily uniform (i.e., cocompact). Also, if every irreducible factor of a Euclidean building has rank at least 3, then, by a theorem of Tits [Tit74], the building is the Bruhat–Tits building associated with a p -adic Lie group. Consequently, in this setting, Theorem 1.1 is equivalent to Corollary 1.3.

1.1. Analogous results. Theorem 1.1 echoes several recent results in the setting of both (coarse) negative curvature and (smooth) nonpositive curvature. For example, consider the situation where Q is a parabolic subgroup in a semisimple real Lie group G , and Γ is a uniform lattice in G ; then the quotient G/Q is a compact smooth manifold, and the induced action $\rho_0 : \Gamma \rightarrow \text{Homeo}(G/Q)$ is smooth, hence bi-Lipschitz. In [KKL19], Kapovich–Kim–Lee showed that any small perturbation of ρ_0 in the space of bi-Lipschitz actions $\text{Hom}(\Gamma, \text{Lip}(G/Q))$ is *conjugate* to ρ_0 . Work of Connell–Islam–Nguyen–Spatzier [CINS23] gives the C^0 version of this result, proving that the action ρ_0 is also topologically stable. The work [CINS23] was inspired by Bowden–Mann [BM20], who proved stability in the rank one setting. In the cases of actions with higher regularity, rigidity was obtained by Kanai [Kan96], Katok–Spatzier [KS97], and Brown–Rodriguez Hetz–Wang [BRW24]. These results can be thought of as analogs of Corollary 1.3 for real Lie groups.

In a different setting, work of Bowden–Mann [BM20], Mann–Manning [MM21], and Mann–Manning–Weisman [MMW22] proves that topological stability holds for the induced action of any Gromov-hyperbolic group Γ on its Gromov boundary $\partial\Gamma$. In particular, the result of Mann–Manning–Weisman implies Theorem 1.1 in the case where the Euclidean building X has rank one (i.e. when X is a tree with no leaves). Mann–Manning–Weisman also proved a version of topological stability for boundary actions of *relatively* hyperbolic groups [MMW24]; this implies in particular that if G is the isometry group of a rank-one symmetric space X of noncompact type, and Γ is a nonuniform lattice in G , the induced action of Γ on the visual boundary ∂X is “relatively topologically stable.” No C^0 stability result, relative or otherwise, is currently known for boundary actions of nonuniform lattices acting on either higher-rank symmetric spaces or higher-rank Euclidean buildings.

1.2. Outline of the paper. Our main theorem relies on a general criterion for topological stability, which we state and prove in Section 2. The criterion is based simultaneously on the notion of a “meandering-hyperbolic” group action developed in [KKL19] and the dynamical coding for boundary actions of arbitrary hyperbolic groups considered in [MMW22]. The rough idea of our criterion is as follows: we suppose that it is possible to construct a pair of “locally finite” combinatorial codings $\mathcal{G}_1, \mathcal{G}_2$ for the action by homeomorphisms of some finitely generated group Γ on a compact metrizable space M . Each coding takes the form of a finite directed graph, whose vertices are labeled by open subsets of M and whose edges are labeled by elements of Γ ; if there is an edge e between two vertices, then the element labeling e takes one of the vertex sets inside the other. A point $z \in M$ is “coded” by $\mathcal{G}_i$ if one can follow an infinite-length path in $\mathcal{G}_i$ to yield an infinite intersection of sets containing z . As in the “meandering-hyperbolic” group actions appearing in [KKL19], we suppose that $\mathcal{G}_1$ encodes every point in M , and that $\mathcal{G}_2$ can be used to “interpolate” between any two $\mathcal{G}_1$ -codings of the same point; then we use arguments inspired by techniques in [KKL19], [MMW22] and [MMW24] to show that the codings can be used to recover the data of the Γ -action on M , and that they are stable under C^0 perturbations.

The rest of the paper is concerned with constructing the codings $\mathcal{G}_1$ and $\mathcal{G}_2$ for the standard actions of a group Γ on the flag boundaries of a Euclidean building X . In Section 3 we review

necessary background on the structure of Euclidean buildings and their flag boundaries, and in Sections 4 and 5 we carry out the actual construction. This is the most technical part of the paper, and involves careful study of the interplay between the geometry of the building X and the dynamics of the action on its flag boundaries. A key step is an estimate (proved in Section 4) which directly relates the expansivity properties of the action on flags to the metric geometry of the action on a Euclidean building. Throughout our arguments, we rely heavily on the *higher-rank Morse lemma* of Kapovich–Leeb–Porti [KLP18], which allows us to assert that certain nice quasi-geodesic rays in X stay within bounded distance of *Weyl cones*—convex subsets of X that play a role similar to that of geodesic rays in the rank-one setting.

In Section 6 we prove that the codings from the preceding sections satisfy the hypotheses of the general stability theorem in Section 2, which completes the proof of Theorem 1.1. The main difficulty here is verifying the interpolation property mentioned previously, which involves further application of tools introduced by Kapovich–Leeb–Porti [KLP18]. Finally, in Section 7, we construct the counterexample alluded to in Remark 1.2, showing that the semi-conjugacies in Theorem 1.1 cannot in general be improved to conjugacies.

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2. STABILITY THROUGH AUTOMATA

Throughout this section, we let Γ be a group and let M be a compact metrizable space. Our goal is to give a general criterion which guarantees that some action $\rho : \Gamma \rightarrow \text{Homeo}(M)$ is topologically stable. This requires some initial technical setup.

2.1. Point coders. The basic idea behind the stability criterion is to encode the dynamics of an action $\rho : \Gamma \rightarrow \text{Homeo}(M)$ with a finite combinatorial object. In the following we review one possible formalism (originally developed in the appendix of [MMW24]) for such a “coding”.

Definition 2.1. Let $\rho : \Gamma \rightarrow \text{Homeo}(M)$ be an action. A *finitary ρ -coder* is a finite directed graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ with vertex labels $\{W(v) : v \in \mathcal{V}\}$ and edge labels $\{\alpha(e) : e \in \mathcal{E}\}$ satisfying the following conditions:

- (1) Each vertex label $W(v)$ is an open subset of M .
- (2) Each edge label $\alpha(e)$ is an element of Γ .
- (3) Whenever there is an edge from z_1 to z_2 labeled by α , there is an inclusion

$$\overline{\rho(\alpha)W(z_2)} \subset W(z_1).$$

The paths in a ρ -coder $\mathcal{G}$ “encode” points in M as follows. Suppose that $(e_n)_{n=1}^\infty$ is a sequence of edges giving an infinite path in a ρ -coder $\mathcal{G}$, and $(z_n)_{n=1}^\infty$ is the corresponding sequence of terminal vertices. Then, the third condition above inductively implies that

$$\rho(\alpha(e_1) \cdots \alpha(e_n))W(z_n)$$

forms a decreasing sequence, and that the intersection of all of the sets in this sequence agrees with

$$\bigcap_{n=1}^\infty \rho(\alpha(e_1) \cdots \alpha(e_n))\overline{W(z_n)}.$$

Since M is compact, this intersection is nonempty, and we think of any point in the intersection as being “coded” by this path in $\mathcal{G}$. If the intersection happens to be a singleton $\{p\}$, then it is useful

to think of the sequence of group elements

$$g_n := \alpha(e_1) \cdots \alpha(e_n)$$

as “converging towards” p . Since we want to use codings to study group actions, it is also useful to allow these codings to “start” at any element in Γ as they converge towards a point in M . We formalize all of this as follows.

Definition 2.2. Let $\mathcal{G}$ be a finitary point coder. A $\mathcal{G}$ -coding is a pair $\mathbf{c} = (g_0, (e_n)_{n=1}^\infty)$, where $g_0 \in \Gamma$ and $(e_n)_{n=1}^\infty$ is an infinite edge path in $\mathcal{G}$. The sequence of group elements

$$g_k := g_0 \alpha(e_1) \cdots \alpha(e_k)$$

is called the *path sequence* associated to $\mathbf{c}$, and the sequence of terminal vertices in the path $(e_n)_{n=1}^\infty$ is called the *terminal vertex sequence* of $\mathbf{c}$. The group element g_0 is called the *initial point* of the coding; we also say that $\mathbf{c}$ *starts at* g_0 .

When $p \in M$, we say that a $\mathcal{G}$ -coding $\mathbf{c}$ with terminal vertex sequence $(z_n)_{n=1}^\infty$ and path sequence $(g_n)_{n=0}^\infty$ is a $(\mathcal{G}, \rho)$ -coding of p if p lies in the intersection

$$(2.1) \quad \bigcap_{n=1}^{\infty} \rho(g_n)W(z_n).$$

If the action ρ is understood from context, we might just say that $\mathbf{c}$ is a $\mathcal{G}$ -coding of p .

In general, a $\mathcal{G}$ -coding $\mathbf{c}$ may not be a $(\mathcal{G}, \rho)$ -coding for a *unique* point $p \in M$. We introduce additional terminology to describe when this occurs:

Definition 2.3. If a $\mathcal{G}$ -coding $\mathbf{c}$ codes a unique point $p \in M$ (in other words, if the intersection in (2.1) is a singleton), then we say that $\mathbf{c}$ is ρ -contracting. We say that a finitary point coder $\mathcal{G}$ is ρ -contracting if every $\mathcal{G}$ -coding is ρ -contracting.

Remark 2.4. The original definition of point coders in the appendix of [MMW24] requires all point coders to be ρ -contracting, but it is occasionally convenient to relax this condition.

The following result, proved in [MMW24], gives a condition ensuring that distinct codings of a common point (possibly arising from distinct coders) are “compatible.”

Lemma 2.5 ([MMW24, Lemma A.3]: close codings are compatible). *For any finite subset $F \subset \Gamma$ and any ρ -contracting point coders $\mathcal{G}, \mathcal{G}'$, there is a number $N = N(\mathcal{G}, \mathcal{G}', F)$ satisfying the following. Suppose that $\mathbf{c}$ is a $\mathcal{G}'$ -coding of p with path sequence $(g_k)_{k=0}^\infty$ and terminal vertex sequence $(z_k)_{k=1}^\infty$, and $\mathbf{d}$ is a $\mathcal{G}$ -coding of p with path sequence $(h_k)_{k=0}^\infty$ and terminal vertex sequence $(y_k)_{k=1}^\infty$. Then, for any indices $m, n \in \mathbb{N}$ satisfying $g_n^{-1}h_m \in F$, we have*

$$(2.2) \quad \rho(g_{n+N})\overline{W(z_{n+N})} \subset \rho(h_m)W(y_m).$$

Motivated by the statement of this lemma, we introduce the following terminology:

Definition 2.6. Fix ρ -coders $\mathcal{G}, \mathcal{H}$ (not necessarily distinct) and a finite subset $F \subset \Gamma$. We say that a $\mathcal{G}$ -coding $\mathbf{c}$ with path sequence $(g_k)_{k=0}^\infty$ and an $\mathcal{H}$ -coding $\mathbf{d}$ with path sequence $(h_j)_{j=0}^\infty$ are F -close if there are arbitrarily large indices m, n such that $g_n^{-1}h_m \in F$.

Remark 2.7. In the definition above, we really mean that *both* indices m and n can be taken arbitrarily large, i.e. that for any $N \in \mathbb{N}$, there is a choice of indices m, n with $g_n^{-1}h_m \in F$ and both $m > N$ and $n > N$. However, if $\mathcal{G}$ and $\mathcal{H}$ are both ρ -contracting and M contains no isolated points, this is the same as asking for infinitely many distinct pairs of indices $(m, n) \in \mathbb{N}^2$ such that $g_n^{-1}h_m \in F$. Indeed, if $m \rightarrow \infty$ and $g_n^{-1}h_m \in F$ for bounded n , then it follows that a subsequence

of h_m is constant and therefore (since the $\rho(h_m)W(y_m)$ are nested) $\rho(h_m)W(y_m)$ is eventually constant. If M contains no isolated points, this is impossible since the nested intersection

$$\bigcap_{m=1}^{\infty} \rho(h_m)W(y_m) = \bigcap_{m=1}^{\infty} \rho(h_m)\overline{W(y_m)}$$

is a singleton while the left hand side equals $\rho(h_m)W(y_m)$, an open set, for a sufficiently large m .

2.2. Generating point coders. Typically one constructs combinatorial codings for an action $\rho : \Gamma \rightarrow \text{Homeo}(M)$ via some kind of “expansive data” for ρ . In this paper, we express this idea through the notion of a *proto-coder*, which *generates* the codings described above.

Definition 2.8. A *proto-coder* $\mathcal{S}$ consists of the following data:

- a finite index set Z ,
- a pair of open coverings $\{W(z) : z \in Z\}, \{U(z) : z \in Z\}$ indexed by Z , satisfying $U(z) \subset W(z)$ for each $z \in Z$, and
- a choice of group element $\alpha(z) \in \Gamma$ for each $z \in Z$.

We say that the proto-coder $\mathcal{S}$ is *adapted to* ρ if it satisfies the following condition: for each pair $y, z \in Z$, if the intersection $\rho(\alpha(z))^{-1}U(z) \cap U(y)$ is nonempty then we have a proper inclusion $\overline{W(y)} \subset \rho(\alpha(z))^{-1}\overline{W(z)}$.

Definition 2.9. When $\mathcal{S}$ is a proto-coder adapted to ρ , we say that $\mathcal{S}$ is ρ -*stable* if, for every pair $y, z \in Z$, we have

$$\rho(\alpha(z))^{-1}U(z) \cap U(y) = \emptyset \iff \rho(\alpha(z))^{-1}\overline{U(z)} \cap \overline{U(y)} = \emptyset.$$

If the action ρ is understood from context, we will just say that $\mathcal{S}$ is *stable*.

2.2.1. Constructing point coders from proto-coders. Any proto-coder $\mathcal{S}$ adapted to ρ determines a ρ -coder $\mathcal{G}$.

Definition 2.10. Let $\mathcal{S}$ be a proto-coder adapted to ρ . The ρ -coder $\mathcal{G}(\mathcal{S}, \rho)$ generated by $\mathcal{S}$ and ρ is determined as follows:

- The vertex set of $\mathcal{G}(\mathcal{S}, \rho)$ is Z , the common index set of $\mathcal{S}$.
- The vertex labels of $\mathcal{G}(\mathcal{S}, \rho)$ are given by the sets $W(z)$ for each $z \in Z$.
- There is an edge e from z to y in $\mathcal{G}(\mathcal{S}, \rho)$ if and only if the set $\rho(\alpha(z))^{-1}U(z) \cap U(y)$ is nonempty. The label of this edge is $\alpha(z)$.

It is easy to check that this actually defines a finitary ρ -coder. The proof of the following lemma is also straightforward, and can already be found in the proof of [MMW22, Lemma 2.10].

Lemma 2.11. *Let $\mathcal{G} = \mathcal{G}(\mathcal{S}, \rho)$ be the point coder generated by $\mathcal{S}$ and ρ . Then every $p \in M$ has a $\mathcal{G}$ -coding starting from the identity element.*

Proof. Fix $p \in M$. We wish to find an infinite edge path $(e_n)_{n=1}^{\infty}$ such that

$$p \in \rho(\alpha(e_1) \cdots \alpha(e_n))W((z_n))$$

for all n , where z_n is the terminal vertex of the edge z_n . Equivalently, we want to find an infinite vertex path $(z_n)_{n=1}^{\infty}$ so that

$$p \in \rho(\alpha(z_1) \cdots \alpha(z_{n-1}))W(z_n)$$

for all n . In fact, we will construct a sequence so that

$$(2.3) \quad p \in \rho(\alpha(z_1) \cdots \alpha(z_{n-1}))U(z_n)$$

for all n ; since $U(z_n) \subset W(z_n)$, this implies the previous condition.

Since $\{U(z) : z \in Z\}$ is a covering, we can find $z_1 \in Z$ so that $p \in U(z_1)$, so (2.3) is satisfied for $n = 1$. Now assume inductively that we have defined a vertex path $z_1, \dots, z_n$ in $\mathcal{G}$ so that (2.3) is satisfied up to n . Since the family of sets $\{U(z)\}$ covers M , there is some vertex z_{n+1} so that

$$\rho(\alpha(z_1) \cdots \alpha(z_n))^{-1}p \in U(z_{n+1}).$$

From (2.3), the point $\rho(\alpha(z_1) \cdots \alpha(z_n))^{-1}p$ also lies in $\rho(\alpha(z_n))^{-1}U(z_n)$. Thus $\rho(\alpha(z_n))^{-1}U(z_n) \cap U(z_{n+1}) \neq \emptyset$. So, there is an edge from z_n to z_{n+1} , which gives a continuation of the vertex path which still satisfies (2.3). $\square$

2.3. Stability criterion. We can now state the main result of the section:

Theorem 2.12. *Let S be a finite generating set for Γ , and let $\rho_0 : \Gamma \rightarrow \text{Homeo}(M)$ be an action. Let $\mathcal{S}, \mathcal{S}'$ be proto-coders adapted to ρ_0 , and suppose that all of the following conditions are satisfied:*

- (P1) *Both proto-coders $\mathcal{S}$ and $\mathcal{S}'$ are ρ_0 -stable (as in Definition 2.9).*
- (P2) *The point coders $\mathcal{G} = \mathcal{G}(\mathcal{S}, \rho_0)$ and $\mathcal{G}' = \mathcal{G}(\mathcal{S}', \rho_0)$ generated by $\mathcal{S}, \mathcal{S}'$ and ρ_0 are both ρ_0 -contracting (as in Definition 2.3).*
- (P3) *For every pair of distinct points $p, p' \in M$, there exists a group element $\gamma \in \Gamma$ so that for any vertices $z, z' \in Z(\mathcal{S})$ satisfying $\rho_0(\gamma)p \in U(z)$ and $\rho_0(\gamma)p' \in U(z')$, we have $W(z) \cap W(z') = \emptyset$.*
- (P4) *There is a finite subset $F \subset \Gamma$ satisfying the following: suppose that $\mathbf{c}_1$ and $\mathbf{c}_2$ are two $(\mathcal{G}, \rho_0)$ -codings of the same point $p \in M$, and that the initial points of $\mathbf{c}_1$ and $\mathbf{c}_2$ both lie in $S \cup \{\text{id}\}$. Then, there is a $\mathcal{G}'$ -coding of p starting at id which is F -close to both $\mathbf{c}_1$ and $\mathbf{c}_2$.*

Then ρ_0 is topologically stable.

2.4. Proof of Theorem 2.12. The proof of Theorem 2.12 follows the outline of the proof of the main theorem in [MMW22], which essentially shows Theorem 2.12 in the special case where the proto-coders $\mathcal{S}$ and $\mathcal{S}'$ agree, and the hypothesis (P4) is replaced with the (stronger) assumption that all of the $(\mathcal{G}, \rho_0)$ -codings are F -close to each other. The more general (P4) is inspired by the “meandering-hyperbolicity” property for actions appearing in [KKL19].

2.4.1. Conditions on the perturbation. To prove Theorem 2.12, we first explicitly describe a neighborhood $\mathcal{U}$ of ρ_0 in $\text{Hom}(\Gamma, \text{Homeo}(M))$ which will consist of representations semi-conjugate to ρ_0 . We want every representation $\rho \in \mathcal{U}$ to satisfy the conditions (C1), (C2), and (C3) described below. The first two conditions are straightforward to state.

(C1) For every pair $y, z \in Z(\mathcal{S})$, we have

$$\rho(\alpha(z))^{-1}U(z) \cap U(y) = \emptyset \iff \rho_0(\alpha(z))^{-1}U(z) \cap U(y) = \emptyset.$$

(C2) For every edge $z \rightarrow y$ in either $\mathcal{G}$ or $\mathcal{G}'$, we have

$$\rho(\alpha(z))\overline{W(y)} \subset W(z).$$

The third condition needs a little more setup. First, assume that the finite set F in condition (P4) is closed under taking inverses by (if necessary) replacing it with a larger set; this ensures that F -closeness is a symmetric condition. Now, by Lemma 2.5, there is a constant $N \in \mathbb{N}$ satisfying the following. Suppose $\mathbf{c}$ is a $\mathcal{H}$ -coding and $\mathbf{d}$ is an $\mathcal{H}'$ -coding for $\mathcal{H}, \mathcal{H}' \in \{\mathcal{G}, \mathcal{G}'\}$, and that $\mathbf{c}$ and $\mathbf{d}$ respectively have path sequences $(g_k)_{k=0}^\infty, (h_k)_{k=0}^\infty$ and terminal vertex sequences $(z_k)_{k=1}^\infty, (y_k)_{k=1}^\infty$. Then, if $g_n^{-1}h_m \in F$ for some pair of indices $m, n \in \mathbb{N}$, we have

$$(2.4) \quad \rho_0(g_{n+N})\overline{W(z_{n+N})} \subset \rho_0(h_m)W(y_m).$$

We impose the following condition on every $\rho \in \mathcal{U}$:

(C3) Fix N as above, and suppose that $\mathbf{c}$ and $\mathbf{d}$ are codings with path sequences $(g_k)_{k=0}^\infty, (h_k)_{k=0}^\infty$ and terminal vertex sequences $(z_k)_{k=1}^\infty, (y_k)_{k=1}^\infty$ as above. Then, if $g_n^{-1}h_m \in F$ for some pair of indices $m, n \in \mathbb{N}$, we have

$$(2.5) \quad \rho(g_{n+N})\overline{W(z_{n+N})} \subset \rho(h_m)W(y_m).$$

We observe:

Proposition 2.13. *All of the conditions (C1)-(C3) are open in $\text{Hom}(\Gamma, \text{Homeo}(M))$.*

Proof. Conditions (C1) and (C2) each correspond to finitely many open conditions. To see that condition (C3) does also, multiply both sides of (2.4) by $\rho_0(g_n^{-1})$ and both sides of (2.5) by $\rho(g_n)^{-1}$, and notice that there are only finitely many possibilities for $g_n^{-1}g_{n+N}$ and $g_n^{-1}h_m$. $\square$

2.4.2. *Defining the inverse of the semi-conjugacy.* Now, fix $\rho \in U$ satisfying the conditions above. We want to define an equivariant surjective map $\phi : M \rightarrow M$ intertwining the ρ and ρ_0 actions; we will define this map by first specifying its inverse, i.e. by specifying the preimages $\phi^{-1}(p)$ for each $p \in M$. As a first step, we let $\mathcal{C}(\mathcal{G})$ denote the set of all $\mathcal{G}$ -codings, and similarly let $\mathcal{C}(\mathcal{G}')$ denote the set of $\mathcal{G}'$ -codings. There is a map

$$\Psi : \mathcal{C}(\mathcal{G}) \sqcup \mathcal{C}(\mathcal{G}') \rightarrow 2^M$$

defined as follows: for a coding $\mathbf{c} \in \mathcal{C}(\mathcal{G}) \sqcup \mathcal{C}(\mathcal{G}')$ with path sequence $(g_k)_{k=0}^\infty$ and terminal vertex sequence $(z_k)_{k=1}^\infty$, define

$$\Psi(\mathbf{c}) = \bigcap_{n=0}^{\infty} \rho(g_n)W(z_n).$$

Condition (C2) implies that $\Psi(\mathbf{c})$ is a nonempty closed subset of M .

Lemma 2.14. *If two codings $\mathbf{c}$ and $\mathbf{d}$ in $\mathcal{C}(\mathcal{G}) \sqcup \mathcal{C}(\mathcal{G}')$ are F -close, then $\Psi(\mathbf{c}) = \Psi(\mathbf{d})$.*

Proof. We will show that $\Psi(\mathbf{c}) \subseteq \Psi(\mathbf{d})$; the other inclusion is symmetric since we have enlarged F to ensure that F -closeness is symmetric. Let $(g_k)_{k=0}^\infty$ and $(h_k)_{k=0}^\infty$ be the path sequences for the pair of codings, and let $(z_k)_{k=1}^\infty$ and $(y_k)_{k=1}^\infty$ be the terminal vertex sequences for $\mathbf{c}$ and $\mathbf{d}$, respectively. It suffices to show that for infinitely many indices $m \in \mathbb{N}$, we have

$$\bigcap_{n=0}^{\infty} \rho(g_n)W(z_n) \subseteq \rho(h_m)W(y_m).$$

By assumption, for infinitely many m , we can find an index n such that $g_n^{-1}h_m \in F$ (see Remark 2.7). From condition (C3), we have

$$\rho(g_{n+N})W(z_{n+N}) \subset \rho(h_m)W(y_m),$$

and so we get the desired inclusion. $\square$

Corollary 2.15. *Let $\mathbf{c}_1$ and $\mathbf{c}_2$ be two $(\mathcal{G}, \rho_0)$ -codings of the same point $p \in M$ with start points $s_1, s_2 \in S \cup \{\text{id}\}$, respectively. Then $\Psi(\mathbf{c}_1) = \Psi(\mathbf{c}_2)$.*

Proof. By condition (P4), there is a $(\mathcal{G}', \rho_0)$ -coding $\mathbf{d}$ of p , starting at the identity, which is F -close to both $\mathbf{c}_1$ and $\mathbf{c}_2$. Then Lemma 2.14 implies that $\Psi(\mathbf{c}_1) = \Psi(\mathbf{d}) = \Psi(\mathbf{c}_2)$. $\square$

The corollary tells us that we may use Ψ to define a map $\Phi : M \rightarrow 2^M$. Explicitly, if p is any point in M , then $\Phi(p)$ is defined to be $\Psi(\mathbf{c})$ for any $(\mathcal{G}, \rho_0)$ -coding $(\text{id}, \mathbf{c})$ of p starting at the identity. By Lemma 2.11, at least one such $(\mathcal{G}, \rho_0)$ -coding exists, and by Corollary 2.15, the choice of coding does not matter, so the map is well-defined.

We can also verify:

Lemma 2.16. *The map $\Phi : M \rightarrow 2^M$ is (ρ_0, ρ) -equivariant, in the sense that for any $\gamma \in \Gamma$ and any $p \in M$, we have*

$$\Phi(\rho_0(\gamma)p) = \rho(\gamma)\Phi(p).$$

Proof. It suffices to check that the claim holds when γ lies in the finite generating set S , since then we can apply induction. So, assume $\gamma = s \in S$. Note that if

$$\mathbf{c} = (\text{id}, (e_n)_{n=1}^\infty)$$

is a $(\mathcal{G}, \rho_0)$ -coding for some $p \in M$ starting at id , then

$$\mathbf{c}' = (s, (e_n)_{n=1}^\infty)$$

is a $(\mathcal{G}, \rho_0)$ -coding for $\rho_0(s)p$, starting at s . Thus by Corollary 2.15 we have $\Psi(\mathbf{c}') = \Phi(\rho_0(s)p)$. On the other hand, since the path sequence for $\mathbf{c}'$ is an s -translate of the path sequence for $\mathbf{c}$, the definition of Ψ implies that

$$\Psi(\mathbf{c}') = \rho(s)\Psi(\mathbf{c}).$$

Then since $\Psi(\mathbf{c}) = \Phi(p)$, we get

$$\Phi(\rho_0(s)p) = \rho(s)\Phi(p).$$

□

2.4.3. Defining the semi-conjugacy. The next step is to show that the map $\Phi : M \rightarrow 2^M$ actually specifies the fibers of a well-defined map $\phi : M \rightarrow M$. Equivalently, we need to check:

Proposition 2.17. *The sets $\{\Phi(p) : p \in M\}$ give a partition of M , indexed by M .*

Proof. There are two things we need to verify: first, that

$$\bigcup_{p \in M} \Phi(p) = M,$$

and secondly, that if p, p' are distinct points in M , then $\Phi(p) \cap \Phi(p') = \emptyset$.

For the first claim, observe that conditions (C1) and (C2) guarantee that the proto-coder $\mathcal{S}$ is also adapted to the perturbed action ρ . Moreover, the conditions also imply that the ρ -coder $\mathcal{G}(\mathcal{S}, \rho)$ generated by $\mathcal{S}$ and ρ is precisely $\mathcal{G}$. So by Lemma 2.11, every point $p \in M$ has a $(\mathcal{G}, \rho)$ -coding starting at the identity. As $\mathcal{G}$ is ρ_0 -contracting, the corresponding infinite path in the underlying graph for $\mathcal{G}$ specifies a $(\mathcal{G}, \rho_0)$ -coding (starting at the identity) of a point $q \in M$, and from the definition it follows immediately that $p \in \Phi(q)$.

Now we turn to the second claim. Fix $p, p' \in M$ distinct. By condition (P3), there exists some $\gamma \in \Gamma$ so that for any vertices $z, z' \in Z$ satisfying $\rho_0(\gamma)p \in U(z), \rho_0(\gamma)p' \in U(z')$, we have $W(z) \cap W(z') = \emptyset$; moreover, such a pair of vertices z, z' exists because the $U(z)$ sets cover M . The construction in the proof of Lemma 2.11 tells us that there exist $(\mathcal{G}, \rho_0)$ -codings of $\rho_0(\gamma)p$ and $\rho_0(\gamma)p'$, each starting at the identity, whose initial vertices are, respectively, z, z' ; then the definition of Φ implies that

$$\Phi(\rho_0(\gamma)p) \subset W(z), \quad \Phi(\rho_0(\gamma)p') \subset W(z'),$$

so in particular $\Phi(\rho_0(\gamma)p)$ and $\Phi(\rho_0(\gamma)p')$ are disjoint. Then Lemma 2.16 (equivariance) shows that the same is true of $\Phi(p), \Phi(p')$. □

The lemma tells us that we can define a map $\phi : M \rightarrow M$ with fibers given by Φ . That is, for each $p \in M$, we define

$$\phi(p) = q \iff p \in \Phi(q).$$

This map is well-defined on all of M by the above; it is transparently surjective since Φ is a well-defined map on M . Moreover, (ρ_0, ρ) -equivariance of Φ immediately implies that ϕ is (ρ, ρ_0) -equivariant.

So, to finish the proof, we just need to show:

Lemma 2.18. *The map $\phi : M \rightarrow M$ is continuous.*

Proof. We proceed by contradiction. Suppose that there is a sequence of points (p_n) in M such that $p_n \rightarrow p$, but $\phi(p_n)$ does not converge to $\phi(p)$. Then since M is compact, we can extract a subsequence so that $\phi(p_n)$ converges to some $q \neq \phi(p)$. Choose $\gamma \in \Gamma$ as in condition (P3) above, and fix vertices z, z' so that

$$\rho_0(\gamma)\phi(p) \in U(z), \quad \rho_0(\gamma)q \in U(z').$$

We note that (P3) implies $W(z) \cap W(z') = \emptyset$.

On the other hand, there are $(\mathcal{G}, \rho_0)$ -codings of $\rho_0(\gamma)\phi(p)$ and $\rho_0(\gamma)q$ (starting at the identity) with initial vertices z, z' . This implies

$$\Phi(\rho_0(\gamma)\phi(p)) \subset W(z), \quad \Phi(\rho_0(\gamma)q) \subset W(z').$$

Then, by equivariance we see that

$$\Phi(\phi(\rho(\gamma)p)) \subset W(z),$$

and then the definition of ϕ implies that $\rho(\gamma)p \in W(z)$. Furthermore, since $\rho(\gamma)$ is a fixed homeomorphism of M , we know that $\rho(\gamma)p_n$ converges to $\rho(\gamma)p$, and therefore $\rho(\gamma)p_n \in W(z)$ for all sufficiently large n .

On the other hand, since $\phi(p_n)$ converges to q , by equivariance $\phi(\rho(\gamma)p_n) = \rho_0(\gamma)\phi(p_n)$ converges to $\rho_0(\gamma)q$. But then, for all sufficiently large n we must have $\phi(\rho(\gamma)p_n) \in U(z')$. This implies that there is a $(\mathcal{G}, \rho_0)$ -coding of $\phi(\rho(\gamma)p_n)$ (starting at the identity) with the initial vertex z' , implying

$$\rho(\gamma)p_n \in \Phi(\phi(\rho(\gamma)p_n)) \subset W(z').$$

This contradicts $W(z) \cap W(z') = \emptyset$. □

3. BACKGROUND AND NOTATION FOR EUCLIDEAN BUILDINGS

In this section we fix our notation and conventions for Euclidean buildings, and prove a handful of basic geometric results. We mostly adopt the terminology in [KL97], and refer to that paper for both precise definitions of the objects appearing here and a thorough introduction to the CAT(0) viewpoint on Euclidean buildings.

3.1. CAT(0) reminders. Let X be a locally compact CAT(0) space. Then X has a *visual boundary* $\partial_\infty X$ and, at each point $x \in X$, a *space of directions* $\Sigma_x X$. Precisely, $\partial_\infty X$ is the space of geodesic rays in X modulo the relation identifying rays whose image have finite Hausdorff distance, and $\Sigma_x X$ is the space of geodesic segments in X with an endpoint at x , modulo the relation identifying segments $[x, y], [x, z]$ with vanishing *Alexandrov angle* $\angle_x(y, z)$ at x . The Alexandrov angle is itself the limit of the CAT(0) comparison angles $\angle_{\bar{x}}(\bar{y}', \bar{z}')$ as $y', z' \rightarrow x$ along $[x, y]$ and $[x, z]$, and defines a CAT(1) metric on $\Sigma_x X$. The visual boundary may similarly be equipped with the *Tits* or *angle metric*: for points $\xi_1, \xi_2 \in \partial_\infty X$, the angular distance $d_T(\xi_1, \xi_2)$ is the supremum over all basepoints $x \in X$ of the Alexandrov angle between the unique representatives of ξ_1, ξ_2 based at x . It is also possible to equip the visual boundary with the *cone topology*. In general, the topology induced by the angle metric is finer than this topology.

For each $x \in X$, there is a well-defined *logarithm map* $\log_x : X - \{x\} \rightarrow \Sigma_x X$, taking a point $y \in X - \{x\}$ to the class of the segment $[x, y]$. Often we will write $\vec{x}y$ for $\log_x(y)$. The logarithm map extends continuously to a surjective map $\log_x : (\partial_\infty X, d_T) \rightarrow \Sigma_x X$, which is continuous (in fact, 1-Lipschitz) if $\Sigma_x X$ is equipped with the angle metric defined by Alexandrov angles at x .

We frequently need the following basic CAT(0) comparison estimate:

Lemma 3.1. *Let a, b, c be pairwise distinct points in a CAT(0) space X . For any $\alpha \in [0, \pi]$ satisfying $\angle_b(a, c) \geq \alpha$, we have*

$$\frac{d_X(a, c)}{d_X(a, b)} \geq \sin(\alpha).$$

Proof. Consider a CAT(0) comparison triangle $\bar{a}, \bar{b}, \bar{c}$ for a, b, c , with Euclidean comparison angle $\angle_{\bar{b}}(\bar{a}, \bar{c})$ at $\bar{b}$. There are two cases. Suppose first that $\angle_{\bar{b}}(\bar{a}, \bar{c}) \geq \pi/2$. Then the Euclidean law of cosines implies that $d_X(a, c)^2 \geq d_X(a, b)^2 + d_X(b, c)^2$, and in particular $d_X(a, c) \geq d_X(a, b)$; thus the desired inequality holds for any $\alpha \in [0, \pi]$. On the other hand if $\angle_{\bar{b}}(\bar{a}, \bar{c}) \leq \pi/2$, then we have

$$\alpha \leq \angle_b(a, c) \leq \angle_{\bar{b}}(\bar{a}, \bar{c}) \leq \pi/2,$$

hence

$$\sin \alpha \leq \sin \angle_b(a, c) \leq \sin \angle_{\bar{b}}(\bar{a}, \bar{c}).$$

Then the Euclidean law of sines implies

$$\frac{d_X(a, c)}{d_X(a, b)} = \frac{\sin \angle_{\bar{b}}(\bar{a}, \bar{c})}{\sin \angle_{\bar{c}}(\bar{a}, \bar{b})} \geq \sin \angle_{\bar{b}}(\bar{a}, \bar{c}) \geq \sin \alpha.$$

□

3.2. Euclidean and spherical buildings. Now let X be a locally compact Euclidean building, i.e. a proper CAT(0) space locally modeled on a *Euclidean Coxeter complex* $(\tilde{a}_{\text{mod}}, \tilde{W})$. Here $\tilde{a}_{\text{mod}}$ is a *model apartment*, a Euclidean space of dimension $k = \text{rank}(X)$, and $\tilde{W}$ is the *affine Weyl group*, a Coxeter group generated by reflections in $\tilde{a}_{\text{mod}}$, acting with compact quotient on $\tilde{a}_{\text{mod}}$. A *Euclidean Weyl chamber* (or a *chamber* if the context is clear) in $\tilde{a}_{\text{mod}}$ is a fundamental domain for the $\tilde{W}$ -action; similarly a chamber in X is any subset of X isometrically identified with a chamber in the model apartment.

3.2.1. Associated spherical buildings. Both the visual boundary $\partial_{\infty} X$ and each space of directions $\Sigma_x X$ have the structure of a *spherical building* modeled on a *spherical Coxeter complex* (a_{mod}, W) ; here a_{mod} is a model sphere of dimension $\text{rank}(X) - 1$, identified with the boundary of the model apartment $\tilde{a}_{\text{mod}}$, and W is the finite reflection group given by the image of $\tilde{W}$ under the representation $\tilde{W} \rightarrow \text{Isom}(a_{\text{mod}})$. The stabilizer in $\tilde{W}$ of any point in $\tilde{a}_{\text{mod}}$ can be identified with a subgroup of W , and there is always at least one point in $\tilde{a}_{\text{mod}}$ whose stabilizer is precisely W . *We will always choose basepoints in $\tilde{a}_{\text{mod}}$ (and in X) to satisfy this condition.*

The sphere a_{mod} has a simplicial structure. A *spherical Weyl chamber* (or just a *chamber* if the context is clear) in a_{mod} is a fundamental domain for the W -action, i.e. a maximal simplex in a_{mod} ; similarly a chamber in $\partial_{\infty} X$ is a subset isometrically identified with a chamber in a_{mod} . Note that the logarithm map takes apartments in the building $\partial_{\infty} X$ to apartments in $\Sigma_x X$ (not necessarily injectively), and takes chambers in $\partial_{\infty} X$ to chambers in $\Sigma_x X$ (isometrically with respect to the induced Tits and Alexandrov distances).

The quotient $\sigma_{\text{mod}} = a_{\text{mod}}/W$ is called the *model Weyl chamber* for the Coxeter complex (a_{mod}, W) . Any spherical building S modeled on (a_{mod}, W) has a *type map* or *anisotropy map* $\theta : S \rightarrow \sigma_{\text{mod}}$, defined on the point $x \in S$ by choosing an apartment $\mathcal{A} \subset S$ containing x , identifying $\mathcal{A}$ with a_{mod} , and mapping x to its image in the quotient $a_{\text{mod}} \rightarrow a_{\text{mod}}/W$. The map is independent of the choice of apartment $\mathcal{A}$ and the choice of identification of $\mathcal{A}$ with a_{mod} . If x is a point in a Euclidean building X , then the logarithm map $\log_x : \partial_{\infty} X \rightarrow \Sigma_x X$ is *type-preserving*, i.e. it satisfies $\theta(\log_x(\xi)) = \theta(\xi)$.

3.3. Flag spaces at infinity. The main result in this paper concerns actions on *flag spaces*, which are spaces of k -simplices in the boundary of a Euclidean building. To study these spaces we adopt the notation and terminology of Kapovich–Leeb–Porti [KLP18]. Kapovich–Leeb–Porti’s setup allows for a unified study of the flag spaces at the boundary of both Euclidean buildings and Riemannian symmetric spaces, but in this paper we are only concerned with the former.

3.3.1. Cones and sectors. Whenever A is a subset of $\partial_\infty X$, and $o \in X$, then $V(o, A)$ denotes the *cone* consisting of the union of all geodesic rays in X based at o and asymptotic to a point in A . If τ_{mod} is a face of the simplex σ_{mod} , and $\tau \subset \partial_\infty X$ is a simplex isometric to τ_{mod} under θ , then the set $V(o, \tau)$ is called a (Euclidean) *Weyl sector of type τ_{mod}* ; when $\sigma \subset \partial_\infty$ is a *maximal* simplex, then $V(o, \sigma)$ is simply called a (Euclidean) *Weyl sector* (instead of Weyl sector of type σ_{mod}). A Weyl sector is a convex subset of an apartment in X .

Definition 3.2. Let $\tau_{\text{mod}} \subset \sigma_{\text{mod}}$ be a closed subsimplex. The τ_{mod} -*flag space* associated to X is the set

$$M = \{\tau \subset \partial_\infty X \mid \tau \text{ is isometric to } \tau_{\text{mod}} \text{ under the type map } \theta\}.$$

After fixing a basepoint $o \in X$, the set M can be equivariantly identified with the set of Weyl sectors $V(o, \tau)$ of type τ_{mod} , and then endowed with the cone topology. Later we will also define a family of metrizations on M , agreeing with this topology (which is independent of the choice of basepoint). Note that M can also be identified with the set of equivalence classes of Weyl sectors of type τ_{mod} , where two sectors of type τ_{mod} are equivalent if they have finite Hausdorff distance from each other.

3.4. Stars and open stars. We now discuss the simplicial structure on the buildings $\partial_\infty X$ and $\Sigma_x X$ in more detail, continuing to adopt the conventions of [KLP18]. If τ_{mod} is a face of σ_{mod} , the *open star* of τ_{mod} , denoted $\text{ost}(\tau_{\text{mod}})$, is the subset of σ_{mod} consisting of the union of open faces in σ_{mod} whose closure contains τ_{mod} . In particular, $\text{ost}(\sigma_{\text{mod}})$ is equal to $\text{int}(\sigma_{\text{mod}})$. On the other hand, if τ_{mod} is a vertex of σ_{mod} , then $\text{ost}(\tau_{\text{mod}})$ is $\sigma_{\text{mod}} - \tau_{\text{mod}}^{\text{opp}}$, where $\tau_{\text{mod}}^{\text{opp}}$ is the closed face opposite to τ_{mod} . See Figure 1.

For a simplex τ in either $\partial_\infty X$ or $\Sigma_x X$ of type τ_{mod} , the *star* of τ , denoted $\text{st}(\tau)$, is the union of all simplices in $\partial_\infty X$ (or $\Sigma_x X$) containing τ . For any $x \in X$, the logarithm map $\log_x : \partial_\infty X \rightarrow \Sigma_x X$ respects the simplicial structure, meaning it takes stars to stars. In fact we have the following stronger statement:

Lemma 3.3 ([KLP18, Lemma 2.9]). *For every $p \in X$ and every simplex $\tau \subset \partial_\infty X$, we have*

$$\log_p(\text{st}(\tau)) = \text{st}(\log_p(\tau)).$$

The nontrivial part of the lemma is that the logarithm map is surjective onto $\text{st}(\log_p(\tau))$. That is, if $\vec{v}$ is a direction at p lying in a face adjacent to $\log_p(\tau)$, then there is a geodesic ray $[p, \xi)$ in the direction of $\vec{v}$, with $\xi \in \partial_\infty X$ lying in a face adjacent to τ .

3.5. The opposition involution. Fix a chamber σ in the model sphere a_{mod} . The Weyl group W has a unique *longest word* w_0 , taking σ to the unique antipodal chamber $-\sigma \subset a_{\text{mod}}$. Identifying σ with the model chamber σ_{mod} , we obtain an involution

$$\iota : \sigma_{\text{mod}} \rightarrow \sigma_{\text{mod}}$$

defined by $\iota(x) = -w_0 x$ for all $x \in \sigma_{\text{mod}}$. Equivalently, since the action of the Weyl group on any apartment $\mathcal{A}$ leaves θ invariant, one may define $\iota(x)$ to be the type $\theta(x^-)$ for any point x^- such that x, x^- are antipodal points in a common apartment $\mathcal{A}$. Note that the first description of ι shows that ι is an isometry with respect to the Tits distance on σ_{mod} , since both the Weyl group W and the map $x \mapsto -x$ act by isometries on the model apartment a_{mod} .

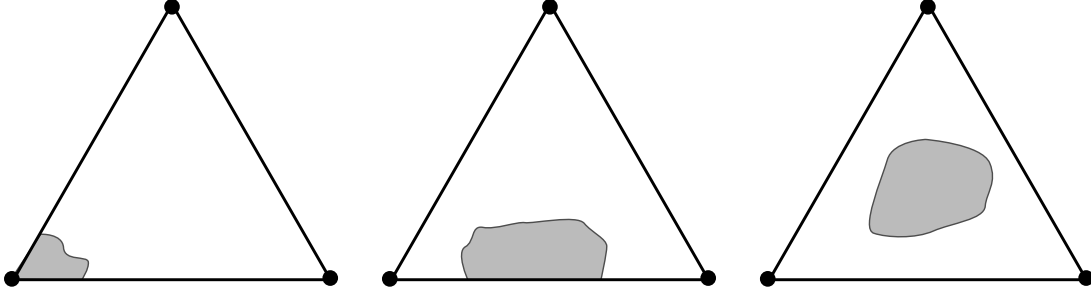


FIGURE 1. Examples of compact subset $\Theta \subset \text{ost}(\tau_{\text{mod}})$ when τ_{mod} is a vertex (left), edge (center), or interior face (right) of a 2-simplex σ_{mod} . Note that the picture is not isometric, since σ_{mod} is a spherical simplex.

We observe:

Proposition 3.4. *If $x, y \in X$ are distinct, then $\theta(\vec{x}\vec{y}) = \iota\theta(\vec{y}\vec{x})$.*

Proof. Consider an apartment $\mathcal{A} \subset X$ containing x and y . The segment $[x, y]$ extends uniquely to a geodesic line $\ell \subset \mathcal{A}$, which can be oriented from x towards y ; since $\log_x$ and $\log_y$ are both type-preserving, the forward ideal endpoint ℓ^+ of ℓ has type $\theta(\vec{x}\vec{y})$, and the backward endpoint ℓ^- has type $\theta(\vec{y}\vec{x})$. Since ℓ^+ and ℓ^- are antipodal points in the spherical apartment $\partial_\infty \mathcal{A} \subset \partial_\infty X$, we have $\iota(\theta(\ell^+)) = \theta(\ell^-)$, hence $\iota\theta(\vec{x}\vec{y}) = \theta(\vec{y}\vec{x})$. $\square$

An easy consequence of the above is the following:

Lemma 3.5. *Let A be an apartment in X , and let $\tau_\pm$ be a pair of antipodal faces in $\partial_\infty A$ (i.e. τ_- consists precisely of the points in $\partial_\infty A$ antipodal to some point in τ_+). Then for any $x, y \in A$, we have $x \in V(y, \tau_+)$ if and only if $y \in V(x, \tau_-)$.*

3.6. Regularity. For $x \in X$, a direction $\vec{v} \in \Sigma_x X$ is called τ_{mod} -regular if $\theta(\vec{v}) \in \text{ost}(\tau_{\text{mod}})$, and a geodesic segment or ray based at x is τ_{mod} -regular if its direction in $\Sigma_x X$ is. A τ_{mod} -regular geodesic ray might not be asymptotic to a unique maximal simplex $\sigma \subset \partial_\infty X$; however, τ_{mod} -regularity ensures that the unique τ_{mod} -face of σ is well-defined, meaning that the ray corresponds to a unique point in τ_{mod} -flag space.

3.6.1. Uniform regularity. A sequence of τ_{mod} -regular rays need not converge to a τ_{mod} -regular ray, because $\text{ost}(\tau_{\text{mod}})$ is not compact unless X has rank one. To guarantee good convergence properties, we ask for our regular rays to be *uniformly regular*. Precisely, fix a compact subset $\Theta \subset \text{ost}(\tau_{\text{mod}})$; a direction $\vec{v}$ is Θ -regular if $\theta(\vec{v}) \in \Theta$, and similarly for rays and segments. The Θ -star of τ is the set

$$\text{st}_\Theta(\tau) = \text{st}(\tau) \cap \theta^{-1}(\Theta).$$

A sequence (x_n) is called Θ -regular if for every $m < n$ we have $\theta(x_n \vec{x}_m) \in \Theta$. The sequence is *uniformly τ_{mod} -regular* if it is Θ -regular for some compact $\Theta \subset \text{ost}(\tau_{\text{mod}})$.

Remark 3.6. There are slightly different conventions for Θ -regularity of a sequence (x_n) as above. One could also ask for such a sequence to satisfy the strictly weaker condition $\theta(x_0 \vec{x}_n) \in \Theta$ for all $n > 0$. In [KLP18], this weaker condition is called “asymptotic uniform regularity”, but we warn the reader that its symmetric-space analog is simply called “uniform regularity” in [KLP17], [KL18].

The following lemma explains why uniformly regular sequences have good convergence properties:

Lemma 3.7. *For any closed set $\Theta \subseteq \sigma_{\text{mod}}$ and any τ_{mod} -flag τ , the Θ -star $\text{st}_{\Theta}(\tau)$ is a closed subset of $\partial_{\infty}X$ with respect to either the angle metric or the cone topology. In particular, if X is locally compact, then $\text{st}_{\Theta}(\tau)$ is compact.*

Proof. We only need to consider the cone topology, since the topology induced by the angle metric is finer. Let $o \in X$ be a basepoint. Note that, if ξ_n is a sequence in $\partial_{\infty}X$ converging to ξ in $\partial_{\infty}X$, then the Alexandrov angles $\angle_o(\xi_n, \xi)$ converge to zero. Thus, since the map $\theta : \partial_{\infty}X \rightarrow \sigma_{\text{mod}}$ factors through the logarithm map $\log_o : \partial_{\infty}X \rightarrow \Sigma_oX$, and θ is 1-Lipschitz on Σ_oX , θ is continuous as a map on $\partial_{\infty}X$ equipped with the cone topology. So, we can finish the proof of the lemma by observing that $\text{st}(\tau)$ is a closed subset of $\partial_{\infty}X$ (with respect to the cone topology); this holds because any limit of spherical Weyl chambers containing τ is also a spherical Weyl chamber containing τ . $\square$

Regularity also allows us to separate both the stars over τ_{mod} -flags and the cones asymptotic to those stars. Precisely, we have the following.

Lemma 3.8 (Separated stars; see [KLP18, Lemma 2.4]). *Let $\Theta \in \text{ost}(\tau_{\text{mod}})$ be compact. If z, z' are distinct τ_{mod} -flags, then for any $\xi \in \text{st}_{\Theta}(z)$ and $\xi' \in \text{st}(z')$, we have*

$$\angle(\xi, \xi') \geq d_{\text{Tits}}(\Theta, \sigma_{\text{mod}} - \text{ost}(\tau_{\text{mod}})).$$

Corollary 3.9. *Let $\Theta \subset \text{ost}(\tau_{\text{mod}})$ be compact, let z, z' be τ_{mod} -flags, and let $x, y \in X$. If, for some $D > 0$, the D -neighborhoods of $V(x, \text{st}_{\Theta}(z))$ and $V(y, \text{st}(z'))$ have infinite-diameter intersection, then $z = z'$.*

Proof. Suppose that the D -neighborhoods of the cones $V(x, \text{st}_{\Theta}(z))$ and $V(y, \text{st}(z'))$ have infinite-diameter intersection, and let (x_n) be a sequence in this intersection whose distance to both x and y tends to infinity. Up to subsequence, x_n converges to a point $\xi \in \partial_{\infty}X$, which must lie in both $\text{st}_{\Theta}(z)$ and $\text{st}(z')$. Thus $z = z'$ by Lemma 3.8. $\square$

This corollary allows us to define the notion of the τ_{mod} -limit of a sequence lying in a neighborhood of a Θ -cone.

Definition 3.10. Let Θ be a compact subset of $\text{ost}(\tau_{\text{mod}})$, and let (x_n) be an unbounded sequence in X , lying in a bounded neighborhood of some cone $V(o, \text{st}_{\Theta}(\tau))$. Then τ is the τ_{mod} -limit of x_n .

Remark 3.11. It is possible to define τ_{mod} -limits for (subsequences of) arbitrary Θ -regular sequences, which makes the space of τ_{mod} -flags into an actual bordification of the space X .

3.6.2. Stability for uniformly regular sequences. Uniform regularity is also stable under perturbation in the following sense: if (x_n) is a Θ -regular sequence, and $\Theta' \subset \sigma_{\text{mod}}$ is a compact set containing Θ in its interior, then any sequence (x'_n) sufficiently close to (x_n) will be Θ' -regular. This is a direct consequence of the following estimate:

Proposition 3.12. *Let x_1, x_2, y_1, y_2 be points in X satisfying $d_X(x_1, x_2) \leq D$, $d_X(y_1, y_2) \leq D$ and $d_X(x_1, y_1) \geq L, d_X(x_2, y_2) \geq L$. Then*

$$\angle(\theta(x_1 \vec{y}_1), \theta(x_2 \vec{y}_2)) \leq 2 \sin^{-1}(D/L).$$

Proof. We first prove that $\angle(\theta(x_1 \vec{y}_1), \theta(x_1 \vec{y}_2)) \leq \sin^{-1}(D/L)$. Since the map $\theta : \Sigma_x X \rightarrow \sigma_{\text{mod}}$ is 1-Lipschitz, it suffices to bound the Alexandrov angle $\angle(x_1 \vec{y}_1, x_1 \vec{y}_2)$. This is bounded by the CAT(0) comparison angle $\angle_{\overline{x_1}}(\overline{y_1}, \overline{y_2})$, which satisfies

$$\angle_{\overline{x_1}}(\overline{y_1}, \overline{y_2}) \leq \sin^{-1} \left(\frac{d_X(y_1, y_2)}{d_X(x_1, y_1)} \right) \leq \sin^{-1}(D/L).$$

as claimed. An identical argument proves that

$$\angle(\theta(y_2\vec{x}_1), \theta(y_2\vec{x}_2)) \leq \sin^{-1}(D/L).$$

Now, using Proposition 3.4 and the triangle inequality for angles on σ_{mod} , we have

$$\begin{aligned} \angle(\theta(x_1\vec{y}_1), \theta(x_2\vec{y}_2)) &\leq \angle(\theta(x_1\vec{y}_1), \theta(x_1\vec{y}_2)) + \angle(\theta(x_1\vec{y}_2), \theta(x_2\vec{y}_2)) \\ &\leq \angle(\theta(x_1\vec{y}_1), \theta(x_1\vec{y}_2)) + \angle(\iota\theta(y_2\vec{x}_1), \iota\theta(y_2\vec{x}_2)) \end{aligned}$$

Since ι acts by an isometry on σ_{mod} , this is bounded beneath

$$\angle(\theta(x_1\vec{y}_1), \theta(x_1\vec{y}_2)) + \angle(\theta(y_2\vec{x}_1), \theta(y_2\vec{x}_2)) \leq 2 \sin^{-1}(D/L),$$

as desired. $\square$

3.7. Convexity and nesting. Frequently, when considering Θ -regular directions or sequences, we will ask for the compact set Θ to be τ_{mod} -convex. A subset $A \subset \sigma_{\text{mod}}$ is τ_{mod} -convex if its symmetrization $W_{\tau_{\text{mod}}}(A) \subset \text{st}(\sigma_{\text{mod}})$ is a convex subset of a_{mod} ; here, $W_{\tau_{\text{mod}}}$ denotes the subgroup of W fixing a face in a_{mod} of type τ_{mod} pointwise. See Figure 2.

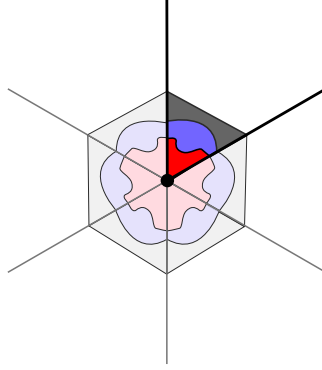


FIGURE 2. Three compact subsets of $\text{ost}(\tau_{\text{mod}})$, when τ_{mod} is a vertex of a 2-simplex σ_{mod} ; the figure depicts a neighborhood of a vertex τ in the model apartment a_{mod} . The innermost set (red) is not convex or τ_{mod} -convex. The middle set (blue) is convex, but not τ_{mod} -convex. The outermost set (gray) is both convex and τ_{mod} -convex.

A τ_{mod} -convex subset gives rise to convex Θ -stars and Θ -cones:

Lemma 3.13 ([KLP18, Proposition 3.16]). *Let τ be a τ_{mod} -flag, and let $\Theta \subset \text{st}(\tau_{\text{mod}})$ be τ_{mod} -convex. Then for any $o \in X$, the cone $V(o, \text{st}_{\Theta}(\tau))$ is a convex subset of X .*

Any compact subset $\Theta \subset \text{ost}(\tau_{\text{mod}})$ can be replaced with the image of its convex hull in $\text{st}(\tau_{\text{mod}})$ under the type map θ , giving rise to a compact τ_{mod} -convex subset of σ_{mod} containing Θ . So in applications, we often assume that Θ is τ_{mod} -convex.

An important consequence of Lemma 3.13 is the following result:

Lemma 3.14 (Nesting lemma for Θ -cones; see [KLP18, Corollary 3.17]). *Let $\Theta \subset \text{st}(\tau_{\text{mod}})$ be τ_{mod} -convex, and let $\tau \in M$ satisfy $\theta(\tau) = \tau_{\text{mod}}$. If $o \in X$ and $p \in V(o, \text{st}_{\Theta}(\tau))$, then $V(p, \text{st}_{\Theta}(\tau)) \subseteq V(o, \text{st}_{\Theta}(\tau))$.*

The nesting property of Θ -cones enables local-to-global arguments in the nonpositively curved space X , since the nested cones prevent “backtracking” for Θ -regular paths.

3.8. The higher-rank Morse lemma. Quasi-geodesics in rank-one symmetric spaces and Euclidean buildings (and more generally, in Gromov-hyperbolic metric spaces) satisfy the *Morse lemma*: any (K, A) -quasi-geodesic stays close to a geodesic ray, with a bound depending only on the constants K, A . Thus a quasi-geodesic sequence in such a space always has a well-defined “endpoint” in the boundary at infinity.

When X is a Euclidean building or Riemannian symmetric space with rank at least two, the Morse lemma fails (in fact it already fails in any isometrically embedded 2-flat in X). However, while *arbitrary* quasi-geodesics may be poorly behaved, work of Kapovich–Leeb–Porti [KLP18] shows that *uniformly* τ_{mod} -regular quasi-geodesics still have well-defined “endpoints” in the space of τ_{mod} -flags in $\partial_\infty X$. Precisely, Kapovich–Leeb–Porti proved the following higher-rank version of the Morse lemma:

Theorem 3.15 (Higher-rank Morse Lemma for Euclidean buildings; see [KLP18, Theorem 1.3]). *Given a compact subset $\Theta \subset \text{ost}(\tau_{\text{mod}})$ and constants $K \geq 1, A \geq 0$, there exists a constant $D \geq 0$ satisfying the following. If (x_n) is any Θ -regular (K, A) -quasi-geodesic sequence in X , then there exists a unique flag $\tau \in M$ so that each x_n lies within a D -neighborhood of $V(x_0, \text{st}(\tau))$.*

One can improve the statement slightly. The corollary below shows that the sequence (x_n) actually must lie within a uniform neighborhood of a Θ' -cone, for some compact $\Theta' \subset \text{ost}(\tau_{\text{mod}})$.

Corollary 3.16. *Let Θ, K, A be as in Theorem 3.15, and let Θ' be any compact subset of $\text{ost}(\tau_{\text{mod}})$ whose interior contains Θ . Then there exists some constant $D > 0$ so that for any $n < m$, the point x_m lies in the D -neighborhood of $V(x_n, \text{st}_{\Theta'}(\tau))$.*

Combining this corollary with Corollary 3.9 shows that uniformly regular quasi-geodesic sequences have well-defined limits in the space of τ_{mod} -flags (see Definition 3.10).

Proof of Corollary 3.16. Fix some $n \in \mathbb{N}$. Since the tail $(x_j)_{j=n}^\infty$ is also a Θ -regular (K, A) -quasi-geodesic sequence, Theorem 3.15 tells us that there is some constant D' and some τ_n such that for all $m > n$, the point x_m lies in the D' -neighborhood of $V(x_n, \text{st}(\tau_n))$. The uniqueness part of Theorem 3.15 implies that actually $\tau_n = \tau_0 = \tau$ for all n .

Choose $\delta > 0$ so that the closed δ -neighborhood of Θ lies in Θ' , and let $L = D'/\sin(\delta/2)$. Fix $m > n$; we wish to show that x_m is uniformly close to the cone $V(x_n, \text{st}(\tau))$. There are two cases: either $m - n > K(L + D' + A)$, or $m - n \leq K(L + D' + A)$.

For the first case, let x'_m be the nearest-point projection of x_m to $V(x_n, \text{st}(\tau))$. In this case, since $(x_j)_{j=0}^\infty$ is a (K, A) -quasi-geodesic sequence, the distance between x_m and x_n is greater than $L + D'$. So, Proposition 3.12 implies that $\angle(\theta(x_n \vec{x}_m), \theta(x_n \vec{x}'_m)) < \delta$; since we have assumed that the sequence (x_n) is Θ -regular, it follows that $\theta(x_n \vec{x}'_m) \in \Theta'$. Thus $x'_m \in V(x_n, \text{st}_{\Theta'}(\tau))$, and we can conclude since x_m is D' -close to x'_m .

For the second case, let M be the smallest integer satisfying $M - n > K(L + D' + A)$. Since $(x_j)_{j=0}^\infty$ is a (K, A) -quasi-geodesic and $n < m < M$, the distance between x_m and x_M is at most $K^2(L + D' + A) + A$, and the first case above shows that x_M is D' -close to a point in $V(x_n, \text{st}_{\Theta'}(\tau))$. So in this case x_m is within distance $K^2(L + D' + A) + A + D'$ of $V(x_n, \text{st}_{\Theta'}(\tau))$. $\square$

Remark 3.17. It is possible to further improve the statement of the corollary as follows: for any Θ, Θ', K, A as in the statement, there is a constant $D > 0$ so that any Θ -regular (K, A) -quasi-geodesic sequence (x_n) is within Hausdorff distance D of another sequence (x'_n) satisfying $x'_m \in V(x'_n, \text{st}_{\Theta'}(\tau))$ for all $n < m$. In the case where X is a symmetric space, rather than a locally compact Euclidean building, a statement similar to this stronger one is proved in [KLP25].

4. EXPANSIVITY

To apply the results of Section 2 towards the main theorem, we need to construct a pair of proto-coders adapted to the action of some cocompact group $\Gamma < \text{Isom}(X)$ on a flag space M for a Euclidean building X . The proto-coders express the fact that the Γ -action on M is “topologically expansive;” to construct the coders, we will use the fact that there is a metrization of M so that the action is “expansive” in a more typical sense. In particular, we will show that for every point $\tau \in M$, there is a neighborhood W of τ , a group element $\alpha \in \Gamma$, and a constant $E > 1$ so that for all $x, y \in W$, we have

$$d(\alpha x, \alpha y) \geq E \cdot d(x, y)$$

for a fixed metric d on M . The purpose of this section is to prove a carefully quantified version of such an expansivity result, stated (in an inverted “contracting” form) as Proposition 4.1 below.

4.1. Metrics on flag spaces. Before stating the result, we identify the metric on M witnessing the expansivity of the action. We actually have a family of metrics d_o on M , varying according to a choice of basepoint o in X . As mentioned previously, we will always assume that o is chosen so that its stabilizer in any apartment is identified with the full spherical Weyl group W .

For each $x \in M$, let $b_x \in \partial_\infty X$ denote the barycenter of the simplex associated to x in $\partial_\infty X$. For $\zeta \in \partial_\infty X$ and $o \in X$, let $[o, \zeta)$ denote the unique geodesic ray in X based at o and asymptotic to ζ . Then, for every $x, y \in M$, define

$$d_o(x, y) = \begin{cases} 0 & \text{if } x = y, \\ 2^{-|[o, b_x) \cap [o, b_y)|} & \text{otherwise.} \end{cases}$$

Here $|[o, \zeta_1) \cap [o, \zeta_2)|$ denotes the length of the (possibly trivial) geodesic segment $[o, \zeta_1) \cap [o, \zeta_2)$. The metric d_o is an *ultrametric*, i.e. for every $x, y, z \in M$, we have

$$d_o(x, z) \leq \max\{d_o(x, y), d_o(y, z)\}.$$

4.2. The contraction estimate. For the estimate below, fix a basepoint $o \in X$ and a model face $\tau_{\text{mod}} \subseteq \sigma_{\text{mod}}$. Let M be the space of τ_{mod} -flags.

Proposition 4.1. *Given $D > 0, E > 1$ and a compact subset $\Theta \subset \text{ost}(\tau_{\text{mod}})$, there exist $L > 0$ (depending only on D, E, Θ) and $\epsilon > 0$ (depending only on D) so that the following holds. If $z \in M$ and $\gamma \in \text{Isom}(X)$ satisfy:*

- (i) $d_X(o, \gamma o) > L$, and
- (ii) γo belongs to the D -neighborhood of the Θ -cone $V(o, \text{st}_\Theta(z))$,

then γ is E^{-1} -contracting on the ball $B(\gamma^{-1}\tau, \epsilon)$ with respect to the metric d_o . That is, $d_o(\gamma x, \gamma y) < \frac{1}{E}d_o(x, y)$ for every $x, y \in B(\gamma^{-1}\tau, \epsilon)$.

Remark 4.2. In the analogous situation where M is a flag boundary for some Riemannian symmetric space, a related expansivity estimate also appears in work of Kapovich–Leeb–Porti (see [KLP17, Theorem 2.41]). Their result, unlike ours, is infinitesimal in nature, and its proof relies on the fact that the flag spaces at the boundary of a symmetric space have the structure of a smooth manifold, with the isometry group acting by diffeomorphisms. In contrast, the statement above is non-infinitesimal; this is partly because the flag space M is no longer a manifold, but it is also because our application requires us to estimate the size of an “expanded” ball. This may not be easy to do using a purely infinitesimal statement.

Intuitively, the estimate in Proposition 4.1 holds because d_o behaves like a visual metric: as we move a basepoint in X “closer” to the ideal endpoints of a pair of geodesic rays, the simplices containing those endpoints look farther apart in the metric determined by the new basepoint. We

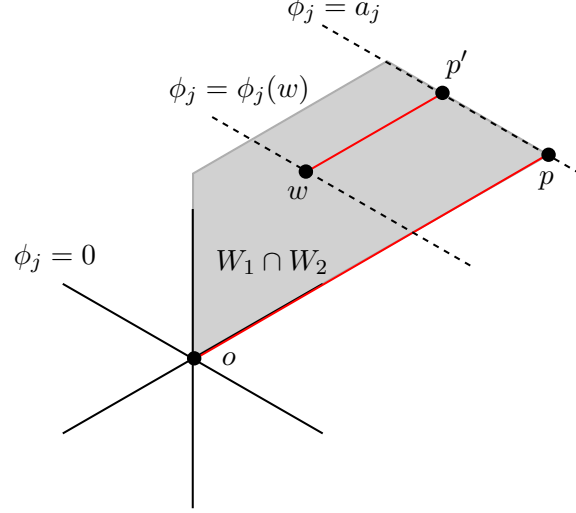


FIGURE 3. Illustration for the proof of Lemma 4.3. In this case τ_{mod} is a vertex of the one-dimensional simplex σ_{mod} .

first show that it is possible to make this intuition precise in the case where one basepoint lies in the intersection of a pair of σ_{mod} -Weyl sectors based at the other. This gives a sharper (but more restrictive) version of Proposition 4.1.

Lemma 4.3 (Change-of-basepoint map in Weyl sectors). *Let $o \in X$, let $x, y \in M$, and let Θ be a compact subset of $\text{ost}(\tau_{\text{mod}})$. Suppose that the geodesics $[o, b_x)$ and $[o, b_y)$ are contained in closed σ_{mod} -Weyl sectors W_1, W_2 with tip at o , and $w \in W_1 \cap W_2$ satisfies $\theta(\vec{o}w) \in \Theta$. There exists a constant $K > 0$, depending only on Θ , so that*

$$d_o(x, y) \leq 2^{-Kd_X(o, w)} d_w(x, y).$$

Proof. The lemma is immediate when $x = y$, so assume $x \neq y$. Let p be the first point where $[o, b_x)$ and $[o, b_y)$ diverge, and let p' be the first point where $[w, b_x)$ and $[w, b_y)$ diverge. Both points p, p' must lie in the boundary of the intersection $W_1 \cap W_2$. This intersection is a convex (not necessarily bounded) polyhedron P in an apartment $\mathcal{A}$, with a vertex at o . Since $\mathcal{A}$ is an apartment, we can identify it with $\mathbb{R}^k$, and assume that o lies at the origin.

Let $W < \tilde{W}$ be the spherical Weyl group (which fixes o), and choose a positive root system $\{\phi_j\} \subset \mathcal{A}^*$ for W . Since the polyhedron P is contained in a Euclidean Weyl sector in $\mathcal{A}$, up to the action of W it is contained in the intersection of half-spaces

$$V = \{x \in \mathcal{A} : \phi_j(x) \geq 0 \text{ for all } \phi_j\}.$$

In fact, P is exactly equal to an intersection of V with finitely many half-spaces of the form $\{x \in \mathcal{A} : \phi_j(x) \leq a_j\}$, with $a_j \geq 0$. The point p must lie on the boundary of one of these half-spaces, meaning there is some positive root ϕ_j such that $\phi_j(p) = a_j < \infty$.

Note that if $p = o$, then $w = o$ and the lemma is immediate, so assume that $p \neq o$. In this case we may choose the positive root ϕ_j so that $\phi_j(p) = a_j > 0$; see Figure 3 for a depiction of the situation. Now, since $p' \in P$, we must have $\phi_j(p') \leq a_j$, hence

$$\phi_j(p) \geq \phi_j(p') = \phi_j(p' - w) + \phi_j(w).$$

The vectors p and $p' - w$ are parallel, lying in the direction of the barycenter of a common simplex τ of type τ_{mod} in $\partial_\infty \mathcal{A}$. This direction is transverse to $\ker(\phi_j)$, so there is a constant $C > 0$

(depending only on τ_{mod}) so that

$$C \cdot \phi_j(p) = d(o, p), \quad C \cdot \phi_j(p' - w) = d(w, p').$$

Thus we have

$$d(o, p) \geq d(w, p') + C \cdot \phi_j(w).$$

Now, since ϕ_j is positive on the direction of the barycenter of $\tau \subset \partial_\infty \mathcal{A}$, it is also positive on vectors in the direction of the open star of τ in $\partial_\infty \mathcal{A}$, and uniformly positive on unit vectors in the direction of any compact subset of $\text{ost}(\tau)$. We have assumed that $\theta(o\vec{w})$ lies in the compact set Θ and that $w \in V$, so this means that there is some uniform constant $K > 0$ so that $C \cdot \phi_j(w) \geq Kd(o, w)$; putting this together with the above, we obtain

$$d(o, p) \geq d(w, p') + Kd(o, w),$$

which gives the desired inequality. $\square$

It is possible to use the estimate above (together with the nestedness property for Θ -cones, see Lemma 3.14) to prove the contraction property in Proposition 4.1 in the special case where the contracting element g takes a basepoint o directly into the cone $V(o, \text{st}(\tau))$. To prove the general contraction estimate, however, we need a version of the nesting lemma which allows us to compare a pair of cones $V(o, \text{st}(\tau))$ and $V(x, \text{st}(\tau))$ when x does *not* lie in $V(o, \text{st}(\tau))$. First, we need a way to estimate the amount of time it takes a geodesic ray asymptotic to a Weyl cone to actually enter the cone. This is closely related to (but not exactly the same as) an estimate in work of Kapovich–Leeb–Porti; compare the next two results to [KLP18, Prop. 3.44 and Cor. 3.45].

Lemma 4.4. *Let α be the angle between the barycenter of τ_{mod} and $\sigma_{\text{mod}} - \text{ost}(\tau_{\text{mod}})$, and let τ be a τ_{mod} -flag. Fix $x \in X$, and let $q \in [x, b_\tau]$ be the point satisfying $[q, b_\tau] = [x, b_\tau] \cap V(o, \text{st}(\tau))$. Then we have*

$$d(x, q) \leq \frac{1}{\sin \alpha} \cdot d(x, V(o, \text{st}(\tau))).$$

Proof. Let p be the nearest-point projection from x to $V(o, \text{st}(\tau))$. By Lemma 3.1, it suffices to prove the inequality $\angle_q(p, x) \geq \alpha$. That is, if we view $q\vec{p}$ and $q\vec{x}$ as points in the spherical building $\Sigma_q X$, we wish to show $\angle(q\vec{p}, q\vec{x}) \geq \alpha$. Thus, since $\theta(q\vec{p}) = \iota(\theta(p\vec{q}))$ is the barycenter of $\iota(\tau_{\text{mod}})$, and ι is an isometry, it suffices to show that $q\vec{p}$ is not contained in $\text{ost}(q\vec{x})$. Suppose for a contradiction that this is not the case, i.e. that $q\vec{p} \in \text{ost}(q\vec{x})$. This implies that any closed spherical Weyl chamber $\sigma \subset \Sigma_q X$ containing $q\vec{p}$ must also contain $q\vec{x}$.

Recall that $V(o, \text{st}(\tau))$ is a union of σ_{mod} -Weyl sectors $V(o, \sigma)$ for all simplices $\sigma \subset \partial_\infty X$ containing τ . Each such sector $V(o, \sigma)$ is itself a union of Euclidean Weyl chambers, i.e. fundamental domains for the action of the affine Weyl group $\tilde{W}$ on an apartment containing $V(o, \sigma)$. Now, since $V(o, \text{st}(\tau))$ is convex, we have $[q, p] \subset V(o, \text{st}(\tau))$, and therefore an initial nontrivial sub-segment of $[q, p]$ is contained in the closure of such a chamber $C \subset V(o, \sigma)$. See Figure 4.

Since C is a Euclidean Weyl chamber whose closure contains q , the image of the logarithm map $\log_q : C - \{q\} \rightarrow \Sigma_q X$ contains a closed maximal simplex in $\Sigma_q X$ containing $q\vec{p}$. Since we have assumed $q\vec{p} \in \text{ost}(q\vec{x})$, we see that $q\vec{x}$ must also be in the image of this logarithm map. But this implies that a non-trivial initial subsegment of $[q, x]$ is contained in C , which is impossible since C is a subset of $V(o, \sigma) \subset V(o, \text{st}(\tau))$ and q is the first point where the geodesic $[x, b_\tau]$ intersects $V(o, \text{st}(\tau))$. $\square$

As a consequence of the previous lemma, we get a similar estimate for the amount of time it takes a geodesic ray to enter certain Θ -cones. This result will be useful later, in Section 6.

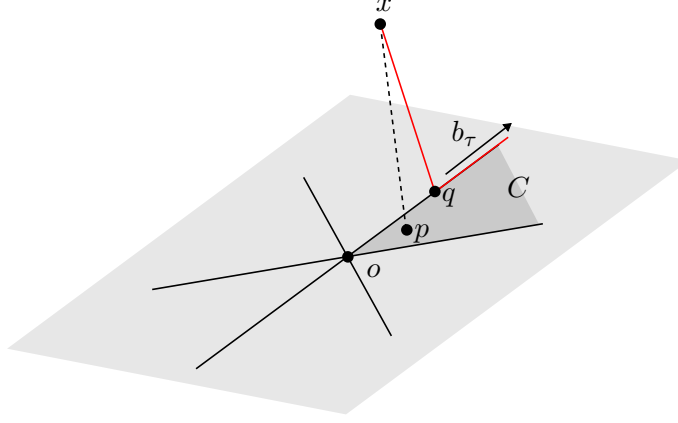


FIGURE 4. Illustration for the proof of Lemma 4.4. As in Figure 3, in this case τ is a vertex in a maximal one-dimensional simplex σ .

Corollary 4.5. *Let α be the angle between the barycenter of τ_{mod} and $\sigma_{\text{mod}} - \text{ost}(\tau_{\text{mod}})$, and let Θ be a subset of σ_{mod} containing a δ -neighborhood of the barycenter of τ_{mod} . Fix a point $x \in X$ and a τ_{mod} -flag τ , and let $q' \in [x, b_\tau)$ be the point satisfying $[q', b_\tau) = [x, b_\tau) \cap V(o, \text{st}_\Theta(\tau))$. Then we have*

$$d_X(x, q') \leq \frac{1}{\sin \alpha} \left(1 + \frac{1}{\sin \delta} \right) d_X(x, o).$$

Note that $\delta \leq \alpha$, so the statement also gives a bound in terms of δ only.

Proof. Let $q \in [x, b_\tau)$ be the point satisfying $[q, b_\tau) = [x, b_\tau) \cap V(o, \text{st}(\tau))$, as in the statement of Lemma 4.4. The proof of Lemma 4.4 shows that $\angle_q(x, o) \geq \alpha$. Thus, by Lemma 3.1, we have $d_X(x, q) \leq \frac{d_X(x, o)}{\sin \alpha}$ and $d_X(o, q) \leq \frac{d_X(x, o)}{\sin \alpha}$.

To get a bound on $d_X(x, q')$, first note that $q' \in [q, b_\tau)$ and $d_X(x, q') = d_X(x, q) + d_X(q, q')$. If $q = q'$ then we are done by the previous estimates, so assume that $q \neq q'$. Since $q \in V(o, \text{st}(\tau))$, there exists a Weyl sector based at o containing q and $[o, b_\tau)$. The ray $[q, b_\tau)$ must also belong to this sector, and be parallel to the ray $[o, b_\tau)$. Therefore, $\angle_{q'}(q, o) = \angle_o(q', b_\tau)$.

On the other hand, since $q \notin V(o, \text{st}_\Theta(\tau))$ and q' is the first point on $[q, b_\tau)$ belonging to $V(o, \text{st}_\Theta(\tau))$, we deduce that $\theta(oq') \in \partial\Theta$. Then, since Θ contains a δ -neighborhood of the barycenter of τ_{mod} , we must have $\angle_o(q', b_\tau) \geq \delta$, hence $\angle_{q'}(q, o) \geq \delta$. Consider the triangle (o, q, q') ; by Lemma 3.1, we get $d_X(q, q') \leq \frac{d_X(o, q)}{\sin \delta}$. Therefore

$$\begin{aligned} d_X(x, q') &= d_X(x, q) + d_X(q, q') \\ &\leq \frac{d_X(x, o)}{\sin \alpha} + \frac{d_X(o, q)}{\sin \delta} \\ &\leq \frac{d_X(x, o)}{\sin \alpha} + \frac{d_X(x, o)}{\sin \alpha \sin \delta}. \end{aligned}$$

□

The following lemma shows that the nesting property for Weyl sectors (see Lemma 3.14) can be transferred to a pair of “nearby” Weyl sectors.

Lemma 4.6. *Fix points $x, y \in X$ and τ_{mod} -simplices $\tau, \tau' \subset \partial_\infty X$, with respective barycenters ξ, ξ' . Suppose that $[y, \xi) \cap [y, \xi')$ contains a nontrivial segment. Then, if the type τ_{mod} -Weyl sector $V(x, \tau)$ contains $[y, \tau)$, the type τ_{mod} -Weyl sector $V(x, \tau')$ contains $[y, \tau')$.*

Proof. Let $\sigma_+ \subset \partial_\infty X$ be a maximal face containing τ . Let $\mathcal{A}$ be an apartment containing $V(x, \sigma_+)$, and let σ_- and τ_- be the antipodal faces to σ_+ and τ in $\partial_\infty \mathcal{A}$ respectively. Since $y \in V(x, \tau)$, by Lemma 3.5 we have $x \in V(y, \tau_-)$.

Now let z be a point in the intersection $[y, \xi] \cap [y, \xi']$ such that the segment $[y, z]$ is nontrivial. We know that $\theta(\vec{y}\vec{z}) = \theta(\xi)$ is the barycenter of τ_{mod} , so let $\tau(\vec{y}\vec{z}) \subset \Sigma_y \mathcal{A}$ denote the type- τ_{mod} simplex whose barycenter is the direction $\vec{y}\vec{z}$, then $\tau(\vec{y}\vec{z}) = \log_y(\tau) = \log_y(\tau')$. By Lemma 3.3, we have

$$(4.1) \quad \log_y \text{st}(\tau) = \log_y \text{st}(\tau') = \text{st}(\tau(y\vec{z})),$$

where $\text{st}(\tau(y\vec{z}))$ is the star in $\Sigma_y X$. Now let $\vec{v}_+ \in \Sigma_y \mathcal{A}$ be the the barycenter of $\log_y(\sigma_+)$, and let $\vec{v}_-$ be its opposite direction in the apartment $\Sigma_y \mathcal{A}$. Since $\vec{v}_+ \in \log_y \text{st}(\tau) = \log_y \text{st}(\tau')$, there exists a σ_{mod} -simplex $\sigma'_+ \subset \text{st}(\tau')$ such that $\theta(\vec{v}_+) \in \theta(\sigma_+)$.

In $\Sigma_y X$, since the barycenters of the maximal simplices $\log_y(\sigma'_+)$ and $\log_y(\sigma_+)$ agree, we have $\log_y(\sigma'_+) = \log_y(\sigma_+)$, and therefore the Weyl sector $V(y, \sigma'_+)$ is opposite to the Weyl sector $V(y, \sigma_-)$. Consequently these Weyl sectors (and their closures) are contained in a common apartment $\mathcal{A}'$, with σ'_+ and σ_- opposite faces in $\partial_\infty \mathcal{A}'$. It follows that the τ_{mod} -face of σ'_+ (namely τ') is opposite to τ_- . Then by Lemma 3.5, since $x \in V(y, \tau_-)$, we have $y \in V(x, \tau')$. Since $V(x, \tau')$ is convex and $\xi' \in \tau'$, this completes the proof. $\square$

Finally we can put everything together and prove the main result of the section.

Proof of Proposition 4.1. Let $x, y \in B(\gamma^{-1}z, \epsilon_0)$, and let $\xi, \eta_1, \eta_2 \in \partial_\infty X$ be the barycenters of the τ_{mod} -simplices z , x , and y respectively. By definition, the rays $[o, \gamma^{-1} \cdot \xi)$ and $[o, \eta_i)$ overlap in a segment of length at least $-\log_2 \epsilon$, for $i = 1, 2$. The rays $[o, \eta_1)$ and $[o, \eta_2)$ overlap in a segment of length equal to $-\log_2 d(\eta_1, \eta_2) \geq -\log_2 \epsilon$. Since γ is an isometry on X , we know that for $i = 1, 2$ the rays $[\gamma \cdot o, \xi)$ and $[\gamma \cdot o, \gamma \cdot \eta_i)$ overlap in a segment of length at least $-\log_2 \epsilon$, and the rays $[\gamma \cdot o, \gamma \cdot \eta_1)$ and $[\gamma \cdot o, \gamma \cdot \eta_2)$ overlap in a segment of length equal to $-\log_2 d(\eta_1, \eta_2) \geq -\log_2 \epsilon$.

The ray $[\gamma \cdot o, \xi)$ is eventually parallel to $[o, \xi)$, so it eventually lies in some Euclidean Weyl sector W containing the τ_{mod} -Weyl sector $V(o, z)$. Note that W has the form $V(o, u) \subset V(o, \text{st}(z))$ for some σ_{mod} -simplex u containing z . Let $w \in [\gamma \cdot o, \xi)$ be the first point where $[\gamma \cdot o, \xi)$ intersects W , so that $[\gamma \cdot o, \xi) \cap W = [w, \xi)$; see Figure 5.

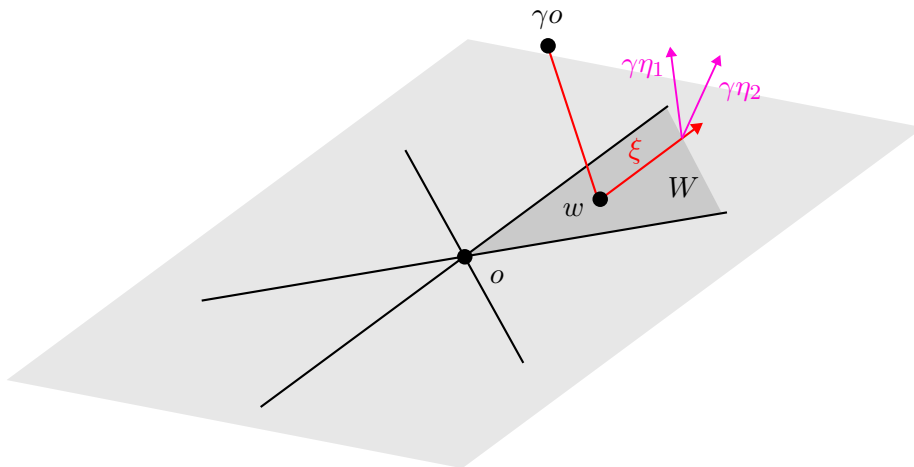


FIGURE 5. Illustration for the proof of Proposition 4.1.

Fix a compact subset $\Theta'' \subset \text{ost}(\tau_{\text{mod}})$ whose interior contains Θ' . We claim that if L is sufficiently large, then $\theta(ow) \in \Theta''$.

To see this, fix a point $w' \in V(o, \text{st}_{\Theta'}(z))$ with $d_X(\gamma \cdot o, w') \leq D$. Then by Lemma 4.4 we have

$$(4.2) \quad d_X(\gamma \cdot o, w) \leq C d_X(\gamma \cdot o, \text{st}(z)) \leq C \cdot d_X(\gamma \cdot o, w') \leq CD,$$

where C is the constant from Lemma 4.4. Thus $d_X(w, w') \leq CD + D$. On the other hand, we have

$$(4.3) \quad \angle_o(w, w') \leq \angle_{\bar{o}}(\bar{w}, \bar{w}') \leq \sin^{-1} \left(\frac{CD + D}{d_X(o, w')} \right) \leq \sin^{-1} \left(\frac{CD + D}{L - D} \right).$$

So, since $\theta(ow') \in \Theta'$ by definition, and $\theta : \Sigma_o X \rightarrow \sigma_{\text{mod}}$ is 1-Lipschitz, if L is large enough then $\theta(ow) \in \Theta''$.

In addition, (4.2) also implies that whenever $-\log_2 \epsilon > CD$, the geodesic segment $[\gamma \cdot o, w]$ is properly contained in $[\gamma \cdot o, \gamma \cdot \eta_1] \cap [\gamma \cdot o, \gamma \cdot \eta_2]$. It follows that

$$\begin{aligned} \log_2 d_{\gamma \cdot o}(\gamma x, \gamma y) &= -|[\gamma \cdot o, \gamma \eta_1] \cap [\gamma \cdot o, \gamma \eta_2]| = -d_X(\gamma \cdot o, w) - |[w, \gamma \eta_1] \cap [w, \gamma \eta_2]| \\ &\geq -CD - |[w, \gamma \eta_1] \cap [w, \gamma \eta_2]| \\ &= -CD + \log_2 d_w(\gamma x, \gamma y). \end{aligned}$$

Furthermore, the geodesic ray $[w, \xi)$ intersects both of the rays $[w, \gamma \cdot \eta_1)$ and $[w, \gamma \cdot \eta_2)$ in nontrivial segments. Then, as a consequence of Lemma 4.6, the Weyl cones $W_1 = V(o, \gamma x)$ and $W_2 = V(o, \gamma y)$ contain the rays $[w, \gamma \cdot \eta_1)$ and $[w, \gamma \cdot \eta_2)$ respectively. Thus, by Lemma 4.3, for some uniform constant K' , depending on Θ'' , we have

$$\begin{aligned} d_o(\gamma x, \gamma y) &\leq 2^{-K' d_X(o, w)} d_w(\gamma x, \gamma y) \\ &\leq 2^{-K' d_X(o, w) + CD} d_{\gamma \cdot o}(\gamma x, \gamma y) \\ &\leq 2^{-K'(L - CD) + CD} d_o(x, y), \end{aligned}$$

where the last inequality follows from the triangle inequality $d_X(o, w) \geq d_X(o, \gamma \cdot o) - d_X(\gamma \cdot o, w) \geq L - CD$. Then, if the constant L satisfies $-K'(L - CD) + CD < -\log_2(E)$, the element γ has the contracting property we desire. $\square$

5. CONSTRUCTING PROTO-CODERS FROM COCOMPACT ACTIONS

In this section we actually set about constructing the proto-coders we need to apply Theorem 2.12. As before, fix a locally compact Euclidean building X . From this point forward, we also assume that $\Gamma < \text{Isom}(X)$ is a discrete group acting properly and cocompactly on X ; thus Γ acts on each flag space associated to a face τ_{mod} of the model simplex σ_{mod} . We fix such a face τ_{mod} , as well as a basepoint $o \in X$. We let $\rho_0 : \Gamma \rightarrow \text{Homeo}(M)$ denote the induced (“standard”) action of Γ on M , although for the most part we still leave this action implicit.

The main result of this section is the following:

Proposition 5.1. *Let Θ be a compact subset of $\text{ost}(\tau_{\text{mod}})$, let $K \geq 1$, and let $A \geq 0$. There exists a ρ_0 -adapted proto-coder $\mathcal{S}$ generating a contracting point coder $\mathcal{G}$ that satisfies both of the following properties:*

- (1) *There is a constant $B > 0$ satisfying the following. If (x_n) is any Θ -regular (K, A) -quasi-geodesic sequence in X , there is a $\mathcal{G}$ -coding $\mathbf{c}$ with path sequence (g_n) such that the Hausdorff distance between (x_n) and $(g_n o)$ is at most B .*
- (2) *There exists a compact set $\Theta' \subset \text{ost}(\tau_{\text{mod}})$ and constants $K' \geq 1, A', D' \geq 0$ satisfying the following. Suppose that $\mathbf{c}$ is a $\mathcal{G}$ -coding of a point $\tau \in M$, with path sequence (g_n) . Then, the sequence $(g_n o)$ is a Θ' -regular (K', A') -quasi-geodesic, lying within a D' -neighborhood of $V(g_0 o, \text{st}_{\Theta'}(\tau))$.*

The two parts of the proposition complement each other: the first part says that any uniformly regular quasi-geodesic in X is uniformly approximated by some coding, and the second part says that every coding determines a uniformly regular quasi-geodesic (but possibly the regularity and quasi-geodesic constants are worse than those from the first part).

5.1. Defining the proto-coder $\mathcal{S}$. For the rest of the section, fix a compact subset $\Theta \subset \text{ost}(\tau_{\text{mod}})$ and quasi-geodesic constants $K \geq 1, A \geq 0$ as in the statement of Proposition 5.1. *For convenience, Θ is chosen so that it contains a small ball, w.r.t. Tits distance, around the barycenter of τ_{mod} .* For the proof of both parts of the proposition, we will need to apply the higher-rank Morse lemma (Theorem 3.15) as a key step, so we also fix an auxiliary τ_{mod} -convex compact set $\Theta' \subset \text{ost}(\tau_{\text{mod}})$ whose interior contains Θ .

Instead of constructing a single $\mathcal{S}$ satisfying the conclusions of Proposition 5.1, we will define a family of proto-coders $\mathcal{S}(D, L, \epsilon)$ depending on real parameters $L, D, \epsilon > 0$. Then we will show that for sufficiently large L, D and sufficiently small $\epsilon > 0$, our construction gives a proto-coder satisfying both of the desired properties.

Step 1: define a pair of open coverings of M . We will employ the contraction estimate from the previous section (Proposition 4.1) to build a pair of nested open coverings of M consisting of “expanding” open sets. Precisely, for each flag $z \in M$, consider the geodesic ray $[o, b_z)$ from o to the barycenter b_z of z . Then, for any parameters $L, D > 0$, let $I_z(D, L)$ be the set of elements $\gamma \in \Gamma$ satisfying both of the following conditions:

- (I) $L \leq d_X(o, \gamma o) \leq 2L$,
- (II) $d_X(\gamma o, V(o, \text{st}_{\Theta'}(z))) < D$.

Although $I_z(D, L)$ depends on L and D , from now on we will omit this from the notation and just write $I_z(D, L) = I_z$.

Observe that I_z is always finite due to the upper bound in (I) and proper discontinuity of the Γ -action on X . Moreover, we have:

Proposition 5.2. *If $L \geq D \geq \text{diam}(X/\Gamma)$, then I_z is nonempty for every $z \in Z$.*

Proof. Let u be any point on the ray $[o, b_z) \subseteq V(o, \text{st}_{\Theta}(z))$ satisfying $d_X(o, u) = L + D$. Then any γ satisfying $d_X(u, \gamma o) \leq D$ verifies both (I) and (II), and such a γ must exist as $\text{diam}(X/\Gamma) \leq D$. $\square$

The above, combined with the contraction estimate from the previous section, implies the following:

Proposition 5.3. *If $D > \text{diam}(X/\Gamma)$, L is sufficiently large (depending only on D) and ϵ is sufficiently small (depending only on D), then each element γ in the nonempty set I_z is $\frac{1}{2}$ -contracting on the ball $B(\gamma^{-1}z, \epsilon)$.*

Proof. Choose L large enough and ϵ small enough as in our contraction estimate (Proposition 4.1); in the hypotheses of the proposition, we take the expansivity constant E to be 2, D to be as above, and the compact set Θ to be our auxiliary compact set $\Theta' \subset \text{ost}(\tau_{\text{mod}})$. Then, if γ satisfies (I) and (II) above, then it also satisfies conditions (i) and (ii) in the hypotheses of Proposition 4.1 with this choice of parameters. $\square$

Now, for each $\epsilon > 0$, and each $\gamma \in I_z$, define the pair of subsets

$$\begin{aligned} U(\gamma, z) &= \gamma B(\gamma^{-1}z, \epsilon/4), \\ W(\gamma, z) &= \gamma B(\gamma^{-1}z, \epsilon/2). \end{aligned}$$

Once again, although $U(\gamma, z)$ and $W(\gamma, z)$ depend on a choice of ϵ , we omit this from the notation. Note that the ultrametric property for the metric d_o on M implies that $U(\gamma, z)$ and $W(\gamma, z)$ are both actually clopen subsets.

Step 2: define $\mathcal{S} = \mathcal{S}(D, L, \epsilon)$ and verify that it is a proto-coder. To define a proto-coder for the ρ_0 -action on M , we need to pick a pair of covers of M , indexed by a common finite set Z , and then assign each element of z to a group element which is “expanding” on the corresponding subsets of the cover. We have defined the sets $U(\gamma, z)$ and $W(\gamma, z)$ so that γ is (metrically) expanding on both of them, so we just need to pick a finite index set $Z \subset \Gamma \times M$. First, for each $z \in M$, consider the clopen subset

$$O(z) = \bigcap_{\gamma \in I_z} U(\gamma, z).$$

Note that $O(z)$ is nonempty as it contains z . The family of sets $\{O(z) : z \in M\}$ also gives a covering of M by open sets, so there is a finite set $F \subset M$ such that $\{O(z) : z \in F\}$ is also a covering. So, define $Z \subset \Gamma \times M$ by

$$Z = \{(\gamma, z) : z \in F, \gamma \in I_z\}.$$

Then Z indexes the pair of open coverings

$$\{W(\gamma, z)\}_{(\gamma, z) \in Z}, \quad \{U(\gamma, z)\}_{(\gamma, z) \in Z}.$$

For each $(\gamma, z) \in Z$, define the “expanding” group element $\alpha(\gamma, z)$ to be γ .

Proposition 5.4. *For any sufficiently large L, D and sufficiently small ϵ , the index set Z , systems of sets $U(\gamma, z), W(\gamma, z)$, and group elements $\alpha(\gamma, z) = \gamma$ define a proto-coder $\mathcal{S}$ adapted to the standard action ρ_0 , as in Definition 2.8.*

Proof. By construction, the sets $U(\gamma, z)$ and $W(\gamma, z)$ give a covering of M indexed by Z , satisfying $U(\gamma, z) \subset W(\gamma, z)$. This means that $\mathcal{S}$ is a proto-coder, so we now just need to verify that it is adapted to the standard action ρ_0 .

Suppose that $(\gamma_1, z_1), (\gamma_2, z_2) \in Z$ satisfy $\alpha(\gamma_1, z_1)^{-1}U(\gamma_1, z_1) \cap U(\gamma_2, z_2) \neq \emptyset$. Then by definition of $U(\gamma_1, z_1)$ and $\alpha(\gamma_1, z_1)$,

$$B(\gamma_1^{-1}z_1, \epsilon/4) \cap U(\gamma_2, z_2) = \gamma_1^{-1}U(\gamma_1, z_1) \cap U(\gamma_2, z_2) \neq \emptyset.$$

Note that γ_2 is a $\frac{1}{2}$ -contracting homeomorphism from a ball of radius $\frac{\epsilon}{4}$ to $U(\gamma_2, z_2)$ and from a ball of radius $\frac{\epsilon}{2}$ to $W(\gamma_2, z_2)$. Thus $U(\gamma_2, z_2)$ and $W(\gamma_2, z_2)$ are contained in balls of radii $\frac{\epsilon}{8}$ and $\frac{\epsilon}{4}$, respectively. So, since $W(\gamma_2, z_2)$ contains at least one point in $B(\gamma_1^{-1}z_1, \epsilon/4)$, the ultrametric property implies

$$\overline{W(\gamma_2, z_2)} = W(\gamma_2, z_2) \subset B(\gamma_1^{-1}z_1, \epsilon/4) \cap W(\gamma_2, z_2).$$

But we also have

$$B(\gamma_1^{-1}z_1, \epsilon/4) \cap W(\gamma_2, z_2) \subset B(\gamma_1^{-1}z_1, \epsilon/4) = \gamma_1^{-1}U(\gamma_1, z_1) \subset \gamma_1^{-1}W(\gamma_1, z_1),$$

implying $\overline{W(\gamma_2, z_2)} \subset \gamma_1^{-1}W(\gamma_1, z_1)$, as required. $\square$

Remark 5.5. We reiterate the order of dependence of the constants in the previous proposition, which ultimately comes from Proposition 5.2 and Proposition 5.3: the conclusion of Proposition 5.4 holds as long as $D > \text{diam}(X/\Gamma)$, L is sufficiently large (depending only on D), and ϵ is sufficiently small (depending only on D). Later we will introduce dependence of ϵ on L , but we will be careful to avoid circularity; L will never depend (directly or indirectly) on ϵ .

From this point forward, we will always assume that L, D and ϵ are chosen so that $\mathcal{S}(D, L, \epsilon)$ actually defines a proto-coder as in the previous proposition. So, we will increase L, D and decrease ϵ as needed throughout the following. We next observe:

Proposition 5.6. *The proto-coder $\mathcal{S}(D, L, \epsilon)$ defined above is stable in the sense of Definition 2.9.*

Proof. The statement follows immediately from the fact that each set $U(\gamma, z)$ is clopen and each $\alpha(\gamma, z)$ is a homeomorphism of M . $\square$

Recall from Section 2.2 that a proto-coder $\mathcal{S}$ adapted to ρ_0 determines a point coder $\mathcal{G}(\mathcal{S}, \rho_0)$. So, from Proposition 5.4, we may define the following.

Definition 5.7. For any D, L, ϵ as above, let $\mathcal{G}(D, L, \epsilon)$ denote the point coder $\mathcal{G}(\mathcal{S}(D, L, \epsilon), \rho_0)$ generated by ρ_0 and the proto-coder $\mathcal{S}(D, L, \epsilon)$. We will refer to a $\mathcal{G}(D, L, \epsilon)$ -coding as a (D, L, ϵ) -coding.

Using the metric contraction of the action of each $\alpha(\gamma, z)$ on $\alpha(\gamma, z)^{-1}W(\gamma, z)$, we can prove:

Proposition 5.8. *The point coder $\mathcal{G}(D, L, \epsilon)$ is ρ_0 -contracting in the sense of Definition 2.3.*

Proof. We defined the sets $W(\gamma, z)$ and chose L, ϵ to ensure that if there is an edge in $\mathcal{G} = \mathcal{G}(D, L, \epsilon)$ from (γ_1, z_1) to (γ_2, z_2) , labeled by $\alpha(\gamma_1, z_1) = \gamma_1$, then γ_1 is $\frac{1}{2}$ -contracting on $\gamma_1^{-1}W(\gamma_1, z_1)$. Therefore γ_1 is also $\frac{1}{2}$ -contracting on $W(\gamma_2, z_2)$. Since we have seen that the proto-coder is adapted to the action of Γ on M , we also know that $\gamma_1 W(\gamma_2, z_2)$ is a subset of $W(\gamma_1, z_1)$.

Consider an infinite vertex path $((\gamma_k, z_k))_{k=1}^\infty$ in $\mathcal{G}$; by definition, the element labeling the edge from (γ_k, z_k) to (γ_{k+1}, z_{k+1}) is γ_k . Applying the observation above inductively, it follows that the set

$$N_k = \alpha_1 \cdots \alpha_k W(\gamma_{k+1}, z_{k+1})$$

has diameter at most $2^{-k} \text{diam } W(\gamma_{k+1}, z_{k+1})$, and that $N_{k+1} \subset N_k$ for all k . Since M is compact, this shows that the intersection of the N_k is a singleton. $\square$

5.1.1. Uniform separation. The next proposition tells us that, if the parameter ϵ is chosen sufficiently small, then condition (P3) in Theorem 2.12 will automatically hold.

Proposition 5.9. *If ϵ is sufficiently small, then for any distinct points $u, u' \in M$, there is a group element $\gamma \in \Gamma$ so that if $\gamma u \in U(\alpha, y)$ and $\gamma u' \in U(\beta, z)$, then*

$$W(\alpha, y) \cap W(\beta, z) = \emptyset.$$

Before we prove the proposition, we recall the following basic fact about cocompact actions on Euclidean buildings. We provide a proof for the reader's convenience.

Lemma 5.10. *Let $M^{(2)}$ denote the space of pairs of distinct points in M . Then, there exists $\epsilon > 0$ so that for any pair $(u, v) \in M^{(2)}$, some element $\gamma \in \Gamma$ satisfies $d_o(\gamma u, \gamma v) > \epsilon$. Consequently, the action of Γ on $M^{(2)}$ is cocompact.*

Proof. Let u, v be a pair of distinct points in M . Then u, v are contained in a common spherical apartment in $\partial_\infty X$ which is the boundary at infinity for a Euclidean apartment $\mathcal{A} \subset X$. Let a be a basepoint in $\mathcal{A}$. Recall that we have assumed that the Γ -action on X is cobounded, so (after translating $\mathcal{A}, a, u$, and v by some element $\gamma \in \Gamma$) we may assume that the distance between a and our fixed basepoint $o \in X$ is at most $\text{diam}(X/\Gamma) < \infty$.

Let $c_u^a : [0, \infty) \rightarrow X$ and $c_v^a : [0, \infty) \rightarrow X$ denote the unit-speed parameterizations of the geodesic rays from a to the barycenters of u, v . The angle between these rays must be one of finitely many nonzero values, so there is some fixed $T > 0$ (independent of u and v) such that

$$d_X(c_u^a(T), c_v^a(T)) \geq 2 \text{diam}(X/\Gamma) + 1.$$

Now, if c_u, c_v denote the unit-speed parameterizations of the geodesic rays from o to the barycenters of u, v , convexity of the distance function implies that

$$d_X(c_u(T), c_u^a(T)) \leq d_X(c_u(0), c_u^a(0)) \leq \text{diam}(X/\Gamma).$$

Similarly we have $d_X(c_v(T), c_v^a(T)) \leq \text{diam}(X/\Gamma)$, and therefore

$$d_X(c_u(T), c_v(T)) \geq 1.$$

By definition this implies $d_X(u, v) \geq \log_2(T)$. $\square$

Proof of Proposition 5.9. Fix $\epsilon > 0$ as in the previous lemma, so that for any $(u, v) \in M^{(2)}$, we can find $\gamma \in \Gamma$ so that $d_o(\gamma u, \gamma v) > \epsilon$. Assume that the proto-coder $\mathcal{S}(D, L, \epsilon)$ is defined using an ϵ which is at least this small. Then, given distinct points $u, v \in M$, fix a γ as above, and suppose that $\gamma u \in U(\alpha, y)$ and $\gamma v \in U(\beta, z)$. Note that the sets $W(\alpha, y)$ and $W(\beta, z)$ both have diameter at most $\epsilon/2$, since they are the images of sets with diameter at most ϵ by $\frac{1}{2}$ -contracting elements. So, as

$$\begin{aligned} \gamma u &\in U(\alpha, y) \subset W(\alpha, y), \\ \gamma v &\in U(\beta, z) \subset W(\beta, z), \end{aligned}$$

if there is some point w in the intersection $W(\alpha, y) \cap W(\beta, z)$, then

$$d_o(\gamma u, \gamma v) \leq d_o(\gamma u, w) + d_o(\gamma v, w) \leq \epsilon.$$

This is a contradiction, so the intersection must be empty. $\square$

5.2. All regular quasi-geodesics are coded. We now turn to the proof of the first condition in Proposition 5.1, which can be viewed as a direct consequence of the following. Recall that we have fixed a compact τ_{mod} -convex subset $\Theta \subset \text{ost}(\tau_{\text{mod}})$ and quasi-geodesic constants $K \geq 1, A \geq 0$.

Proposition 5.11. *For any sufficiently large D, L and any sufficiently small $\epsilon > 0$, there is a constant $B > 0$ satisfying the following. If $(x_n)_{n=0}^\infty$ is a Θ -regular (K, A) -quasi-geodesic sequence in X , then there is a (D, L, ϵ) -coding $\mathbf{c}$ with path sequence $(g_n)_{n=0}^\infty$ such that the Hausdorff distance between $\{x_n\}_{n=0}^\infty$ and $\{g_n o\}_{n=0}^\infty$ is at most B .*

Before proving Proposition 5.11, we need a pair of general lemmas. The first says that any quasi-geodesic has a “reparameterization” which is coarsely proportional to its arc-length.

Lemma 5.12. *Let $K \geq 1$ and $A \geq 0$. For every $L \geq K + A$, there exists $A' \geq 0$ such that every Θ -regular (K, A) -quasi-geodesic (x_n) in X has a subsequence (x_{n_k}) satisfying the following properties:*

- (1) (x_{n_k}) is a Θ -regular quasi-geodesic,
- (2) for every k , we have $L \leq d_X(x_{n_k}, x_{n_{k+1}}) \leq 2L$,
- (3) the sequence (x_n) and subsequence (x_{n_k}) are A' -uniformly close to each other.

Proof. We construct our subsequence term-by-term, making sure we satisfy the second condition each time. Precisely, let $x_{n_1} = x_1$. Then, for all $k > 1$, inductively define n_k to be the smallest index $n_k > n_{k-1}$ so that $d(x_{n_{k-1}}, x_{n_k}) \geq L$. Since $d_X(x_{n_k}, x_{n_{k-1}}) < L$ and (x_n) is (K, A) -quasi-geodesic, it follows from the triangle inequality that $d_X(x_{n_{k-1}}, x_{n_k}) < L + (K + A) \leq 2L$.

Now, since (x_n) is a (K, A) -quasi-geodesic, $K^{-1}(n_{k+1} - n_k) - A \leq d_X(x_{n_k}, x_{n_{k+1}}) \leq K(n_{k+1} - n_k) + A$. It follows that $n_{k+1} - n_k \leq K(2L + A)$. Thus, every element x_n is within distance $A' = K^2(2L + A) + A$ of an element in the subsequence (x_{n_k}) . This shows the third condition. The

first condition also follows: any subsequence of a Θ -regular sequence is automatically Θ -regular, and since $j - k \leq n_j - n_k$ for all $j \leq k$ we have

$$\frac{1}{K}|j - k| - A \leq d_X(x_{n_j}, x_{n_k}) \leq 2L|j - k|$$

for all k, j . □

The second lemma says that the cones over a pair of nearby stars locally agree when they share a basepoint. It is an immediate consequence of a result of Kapovich–Leeb–Porti.

Lemma 5.13 (see [KLP18, Lem. 3.48]). *For any $\ell > 0$, there exists $\epsilon > 0$ such that if $y, z \in M$ satisfy $d_o(y, z) < \epsilon$, then*

$$V(o, \text{st}(y)) \cap B(o, \ell) = V(o, \text{st}(z)) \cap B(o, \ell).$$

Proof of Proposition 5.11. Fix a Θ -regular (K, A) -quasi-geodesic sequence $(x_n)_{n=0}^\infty$ as in the proposition. Recall that we fixed a compact subset $\Theta' \subset \text{ost}(\tau_{\text{mod}})$ whose interior contains Θ . As a consequence of the higher-rank Morse lemma (see Corollary 3.16), there is a constant $D > 0$ (depending only on K, A, Θ, Θ') and some fixed $z \in M$ so that for any $n < m$, the point x_m lies within distance D of the cone $V(x_n, \text{st}_{\Theta'}(z))$.

We can consider a sequence of the form $(g_n o)_{n=0}^\infty$ for $g_n \in \Gamma$, such that the distance between $g_n o$ and x_n is at most $\text{diam}(X/\Gamma)$. After increasing D appropriately (and in particular ensuring that $D > \text{diam}(X/\Gamma)$), it is also true that for any $n < m$, the point $g_m o$ lies in the D -neighborhood of $V(g_n o, \text{st}_{\Theta'}(z))$, because the Hausdorff distance between the Weyl cones $V(x_n, \text{st}_{\Theta'}(z))$ and $V(g_n o, \text{st}_{\Theta'}(z))$ is at most $d_X(x_n, g_n o) \leq \text{diam}(X/\Gamma)$. This is the parameter D we use to define the proto-coder $\mathcal{S}(D, L, \epsilon)$.

Next we fix a constant L , depending on D . There are two lower bounds L needs to satisfy: first, it must be large enough for the conclusion of Proposition 5.4 to hold for any sufficiently small $\epsilon > 0$ (see Remark 5.5). Second, L needs to be large enough for us to replace $(g_n o)$ with a uniformly Hausdorff-close subsequence satisfying

$$(5.1) \quad L \leq d_X(g_n o, g_{n+1} o) \leq 2L$$

for every n . Since $(g_n o)$ is a quasi-geodesic sequence with quasi-geodesic constants depending only on K, A and $\text{diam}(X/\Gamma)$, this is possible due to Lemma 5.12.

Our task is now to check that, if $\epsilon > 0$ is sufficiently small, the sequence $(g_n)_{n=0}^\infty$ is exactly the path sequence for some (D, L, ϵ) -coding $\mathbf{c}$. Since $(g_n o)_{n=0}^\infty$ has uniformly bounded Hausdorff distance to $(x_n)_{n=0}^\infty$, this is sufficient. We will assume throughout that $\epsilon > 0$ is chosen small enough to guarantee (via Proposition 5.4) that $\mathcal{S}(D, L, \epsilon)$ is actually a proto-coder, and thus (D, L, ϵ) -codings actually make sense; we will shrink ϵ as necessarily throughout the following argument.

Precisely, defining $\gamma_n = g_{n-1}^{-1} g_n$ for all $n \geq 1$, we want to show that if $\epsilon < 0$ is sufficiently small, there is a sequence of elements $(z_n)_{n=1}^\infty$ in M so that the sequence $(\gamma_n, z_n)_{n=1}^\infty$ specifies a vertex path in the graph $\mathcal{G} = \mathcal{G}(D, L, \epsilon)$, and therefore defines a (D, L, ϵ) -coding starting at g_0 with path sequence $(g_n)_{n=0}^\infty$.

We start by defining a candidate sequence z_n . Recall that we previously found a finite subset $F \subset M$ so that the F -indexed collection of intersections

$$\left\{ \bigcap_{\gamma \in I_z} U(\gamma, z) : z \in F \right\}$$

gives an open covering of M . In particular, for each $n \geq 1$, there is some $z_n \in F$ so that

$$g_{n-1}^{-1}z \in \bigcap_{\gamma \in I_{z_n}} U(\gamma, z_n).$$

We need to show both that $\gamma_n \in I_{z_n}$ for all $n \geq 1$ (meaning (γ_n, z_n) is in the vertex set Z of $\mathcal{G}$ for all $n \geq 1$) and that there is an edge from (γ_n, z_n) to (γ_{n+1}, z_{n+1}) for all $n \geq 1$.

Note that for any $\gamma \in I_{z_n}$, the set $U(\gamma, z)$ is contained in a d_o -ball of radius $\epsilon/8$, since $\rho(\gamma)$ is $\frac{1}{2}$ -contracting on $\rho(\gamma)^{-1}U(\gamma, z_n) \subset B(\rho(\gamma^{-1})z_n, \epsilon/4)$. So by the ultrametric property we have $d_o(z_n, g_{n-1}^{-1}z) < \epsilon/8$. Thus, by Lemma 5.13, if $\epsilon > 0$ is small enough, we can guarantee that

$$(5.2) \quad V(o, \text{st}_{\Theta'}(z_n)) \cap B(o, 2L + D) = V(o, \text{st}_{\Theta'}(g_{n-1}^{-1}z)) \cap B(o, 2L + D).$$

Now, as $g_n o$ lies within distance D of $V(g_{n-1} o, \text{st}_{\Theta'}(z))$, the point $g_{n-1}^{-1}g_n o = \gamma_n o$ lies within distance D of $V(o, \text{st}_{\Theta'}(g_{n-1}^{-1}z))$. From (5.1) we have

$$L \leq d_X(o, \gamma_n o) \leq 2L.$$

The upper bound, together with (5.2), therefore tells us that $\gamma_n o$ must lie within distance D of $V(o, \text{st}_{\Theta'}(z_n))$. This—combined with the lower bound on $d_X(o, \gamma_n o)$ —is precisely what is needed to ensure that $\gamma_n \in I_{z_n}$ (recall that I_{z_n} is defined by the conditions (I), (II) above).

Finally we note that there is an edge from (γ_n, z_n) to (γ_{n+1}, z_{n+1}) : this occurs exactly when the intersection

$$\rho(\gamma_n)^{-1}U(\gamma_n, z_n) \cap U(\gamma_{n+1}, z_{n+1})$$

is nonempty, and by construction, the point $g_n^{-1}z = \gamma_n^{-1}g_{n-1}^{-1}z$ lies in both sets. $\square$

Remark 5.14. The proof of Proposition 5.11 introduces a dependence of ϵ on L , mentioned previously in Remark 5.5; we will see another such dependence in the sequel.

5.3. Every coding gives a regular quasi-geodesic. Now we wish to prove the second condition in Proposition 5.1. The desired statement is a consequence of the following more technical proposition. Recall that, as part of the definition of the family of proto-coders $\mathcal{S}(D, L, \epsilon)$, we fixed an auxiliary τ_{mod} -convex compact subset $\Theta' \subset \text{ost}(\tau_{\text{mod}})$ whose interior contains Θ .

Proposition 5.15. *Fix any $D > 0$ and a compact subset $\Theta'' \subset \text{ost}(\tau_{\text{mod}})$ whose interior contains Θ' . For any sufficiently large L and any sufficiently small $\epsilon > 0$, there exist $K' \geq 1$, $A', D' \geq 0$ satisfying the following: if $\mathbf{c}$ is a (D, L, ϵ) -coding of a point $z \in M$ with path sequence $(g_n)_{n=0}^\infty$, then the sequence $(g_n o)_{n=0}^\infty$ is a Θ'' -regular (K', A') -quasi-geodesic. Moreover, this quasi-geodesic lies within a D' -neighborhood of $V(g_0 o, \text{st}(z))$.*

The “moreover” part of the proposition is almost a direct consequence of the first part combined with the higher-rank Morse lemma (Theorem 3.15): if the first part of the proposition holds, then the Morse lemma says that the sequence $(g_n o)$ stays close to the cone over *some* star $\text{st}(y)$. However, one still needs to check that, if (g_n) is the path sequence for a coding of z , then $(g_n o)$ actually approaches the specific star $\text{st}(z)$.

The first step in the proof of Proposition 5.15 is the following consequence of uniform contraction:

Lemma 5.16. *There exists a constant B satisfying the following. Given any $R > 0$, if L is sufficiently large (depending on R, D) and ϵ is sufficiently small (depending on L, R, D), and $\mathbf{c}$ is a (D, L, ϵ) -coding with path sequence $(g_n)_{n=0}^\infty$, then for all $n \geq 0$ we have*

$$d_X(g_0 o, g_n o) \geq Rn - B.$$

Since the distance between consecutive points in a coding is uniformly bounded by $2L$, this lemma immediately implies that (D, L, ϵ) -codings determine quasi-geodesic sequences.

Proof. Without loss of generality, assume $g_0 = \text{id}$. Let $E \geq 2$ be arbitrarily chosen. We can choose some $L = L(E)$ at least as large as the value given by Proposition 4.1, with our given value of E and Θ , D chosen as in the proof of Proposition 5.8; then, if $(\gamma_n, z_n)_{n=1}^\infty$ denotes the sequence of vertices in $\mathcal{G}(D, L, \epsilon)$ determining a coding $\mathbf{c}$, our definition of $\mathcal{S}(D, L, \epsilon)$ ensures that each γ_n is an E^{-1} -contracting map from $W(\gamma_{n+1}, z_{n+1})$ to $W(\gamma_n, z_n)$. By the nesting property of vertex sets in the proto-coder, it follows inductively that for all $n \geq 1$, g_n is a E^{-n} -contracting map from $W(\gamma_{n+1}, z_{n+1})$ to $W(\gamma_1, z_1)$.

Let S be a finite generating set for Γ . Each $s \in S$ defines a bi-Lipschitz homeomorphism on M , meaning that there is a constant $0 < C < 1$ so that $d_o(sx, sy) \geq Cd_o(x, y)$ for all $x, y \in M$ and all $s \in S$. Inductively, if some element $h \in \Gamma$ has length at most n (with respect to the word metric on Γ induced by S), we have $d_o(hx, hy) \geq C^n d_o(x, y)$. Consequently, the word-length of g_n is at least $-n \log(E)/\log(C)$. Finally, since Γ acts properly and D -coboundedly on X , the Milnor-Schwartz lemma tells us that the orbit map into X based at o is a quasi-isometric embedding, which gives the desired lower bound if we increase L (hence E) appropriately. $\square$

Our next task is to show that (D, L, ϵ) -codings give uniformly regular sequences. We start with an a priori weaker statement: we prove that the orbits of codings in X can experience at most uniformly linear drift away from the cone over a star.

Lemma 5.17. *For any sufficiently small $\epsilon > 0$ (depending on D, L), if $\mathbf{c}$ is a (D, L, ϵ) -coding of $z \in M$ with path sequence $(g_n)_{n=0}^\infty$, then for all $n \geq 0$ we have*

$$d_X(g_n o, V(g_0 o, \text{st}_{\Theta'}(z))) \leq nD.$$

Note that this lemma also introduces a dependence of ϵ on L (see Remark 5.5).

Proof. Again, without loss of generality, take $g_0 = \text{id}$. Let $(\gamma_n, z_n)_{n=1}^\infty$ denote the sequence of vertices in Z determining the coding $\mathbf{c}$; since the coding starts from the identity, we have $g_n = \gamma_1 \cdots \gamma_n$. By definition, since $\mathbf{c}$ codes z , we have

$$z \in \bigcap_{n=1}^\infty g_{n-1} W(\gamma_n, z_n),$$

and thus for every $n \geq 1$, we have $g_{n-1}^{-1} z \in W(\gamma_n, z_n)$. Now, for every n , we know that $g_{n-1}^{-1} z \in W(\gamma_n, z_n)$, meaning that $d_o(g_{n-1}^{-1} z, z_n) \leq \epsilon/2$. So, by Lemma 5.13, we can choose ϵ small enough so that

$$(5.3) \quad V(o, \text{st}_{\Theta'}(z_n)) \cap B(o, 2L + D) = V(o, \text{st}_{\Theta'}(g_{n-1}^{-1} z)) \cap B(o, 2L + D)$$

holds for all n . Our definition of the proto-coder $\mathcal{S}(D, L, \epsilon)$ ensures that $\gamma_n o$ lies in the D -neighborhood of the cone $V(o, \text{st}_{\Theta'}(z_n))$, so for each n fix a point $p_n \in V(o, \text{st}_{\Theta'}(z_n))$ with $d_X(\gamma_n o, p_n) \leq D$.

We now prove the desired inequality via induction on n . Since $g_0 = \text{id}$, the statement is immediate when $n = 0$, so suppose that $n \geq 1$. Inductively, there is a point $q \in V(o, \text{st}_{\Theta'}(z))$ such that

$$d_X(q, g_{n-1} o) \leq (n-1)D.$$

Now, since $d_X(o, p_n) \leq d_X(o, \gamma_n o) + D \leq 2L + D$, the equality in (5.3) tells us that there is a point $\xi_n \in \text{st}_{\Theta'}(z)$ so that the geodesic ray $[o, g_{n-1}^{-1} \xi_n]$ passes through p_n ; equivalently, the ray $[g_{n-1} o, \xi_n]$ passes through $g_{n-1} p_n$. By convexity, the Hausdorff distance between the rays $[g_{n-1} o, \xi_n]$ and $[q, \xi_n]$ is at most $d_X(q, g_{n-1} o) \leq (n-1)D$. This means that the distance between $g_n o = g_{n-1} \gamma_n o$ and

$[q, \xi_n]$ is at most

$$\begin{aligned} d_X(g_n o, g_{n-1} p_n) + d_X(g_{n-1} p_n, [q, \xi_n]) &= d_X(\gamma_n o, p_n) + d_X(g_{n-1} p_n, [q, \xi_n]) \\ &\leq D + (n-1)D = nD. \end{aligned}$$

Using the nesting property of cones over τ_{mod} -convex stars (Lemma 3.14), since $q \in V(o, \text{st}_{\Theta'}(z))$ and $\xi_n \in \text{st}_{\Theta'}(z)$ we have $[q, \xi_n] \subset V(o, \text{st}_{\Theta'}(z))$, hence $d_X(g_n o, V(o, \text{st}_{\Theta'}(z))) \leq nD$ as required. $\square$

Proof of Proposition 5.15. Let $\delta = \frac{1}{4}d_T(\Theta', \sigma_{\text{mod}} - \text{ost}(\tau_{\text{mod}}))$; in particular this implies that the closed 3δ -neighborhood of Θ' is a compact subset of $\text{ost}(\tau_{\text{mod}})$. Let Θ'' be the δ -neighborhood of Θ' , and let $B > 0$ be the constant from Lemma 5.16. Choose $R > 0$ large enough so that

$$\sin^{-1}\left(\frac{Dn}{Rn - B}\right) < \delta$$

for all $n \geq 1$. Then fix L (depending on R, D) large enough to ensure that the conclusion of Lemma 5.16 holds, and finally fix $\epsilon > 0$ small enough (depending on R, D, L) so that the conclusion of Lemma 5.17 holds.

Fix a (D, L, ϵ) -coding $\mathbf{c}$ for a point $z \in M$, with path sequence $(g_n)_{n=0}^\infty$; as before, we can assume without loss of generality that $g_0 = \text{id}$. Let $x_n = g_n o$. We will show that $\theta(o\vec{x}_n) \in \Theta''$ for all n . By Lemma 5.17, there is a point $p_n \in V(o, \text{st}_{\Theta'}(z))$ with

$$d_X(p_n, x_n) < nD.$$

Consequently the CAT(0) comparison angle $\angle_{\bar{o}}(\overline{p_n}, \overline{x_n})$ is at most

$$\sin^{-1}\left(\frac{d_X(o, x_n)}{d_X(x_n, p_n)}\right) \leq \sin^{-1}\left(\frac{Dn}{Rn - B}\right) < \delta.$$

It follows that the angle between the directions $\theta(o\vec{x}_n)$ and $\theta(o\vec{p}_n)$ is also at most δ , ensuring that $\theta(o\vec{x}_n) \in \Theta''$.

To show that $(x_n)_{n=0}^\infty$ is Θ'' -regular, we need to further show that $\theta(x_i \vec{x}_j) \in \Theta''$ for all $i < j$, but this follows from the above: if the vertex sequence for the coding $\mathbf{c}$ is $(\gamma_n, z_n)_{n=1}^\infty$, then for any $i \geq 0$ the sequence $(\gamma_{i+n}, z_{i+n})_{n=1}^\infty$ also determines a (D, L, ϵ) -coding, with initial point at the identity. The argument above (applied to this coding) shows precisely that for any $j > i$, the geodesic segment from o to $\gamma_{i+1} \cdots \gamma_j o = g_i^{-1} g_j o$ is Θ'' -regular, and therefore so is the geodesic segment from $g_i o$ to $g_j o$.

It remains to show the “moreover” part of the proposition, i.e. that the sequence $(x_n)_{n=0}^\infty$ lies in the D' -neighborhood of the cone $V(o, \text{st}(z))$, for some uniform D' . We have just shown that $(x_n)_{n=0}^\infty$ specifies a Θ'' -regular (K', A') -quasi-geodesic, so the higher rank Morse lemma (Theorem 3.15) provides some uniform D' and some point $z' \in M$ so that x_n lies within distance D' of $V(o, \text{st}(z'))$. So, we only need to check that $z' = z$. The idea is to inspect the convergence of the sequence x_n to a point in $\partial_\infty X$ to show that the angle between two points in $\text{st}_{\Theta''}(z)$ and $\text{st}(z')$ is small, and then apply the separated stars lemma (Lemma 3.8).

Fix a sequence of points $p'_n \in V(o, \text{st}(z'))$ with $d_X(x_n, p'_n) < D'$. Since each cone $V(o, \text{st}(z))$ and $V(o, \text{st}(z'))$ is a union of geodesic rays based at o , for each n we can find points $\xi_n \in \text{st}(z)$ and $\xi'_n \in \text{st}(z')$ so that the geodesic ray $[o, \xi_n]$ passes through p_n , and $[o, \xi'_n]$ passes through p'_n . Up to extracting a subsequence, ξ_n converges to a point $\xi \in \text{st}(z)$ and ξ'_n converges to some $\xi' \in \text{st}(z')$. In fact, since $p_n \in V(o, \text{st}_{\Theta''}(z))$, we know that $\xi_n \in \text{st}_{\Theta''}(z)$, hence $\xi \in \text{st}_{\Theta''}(z)$ by the compactness of Θ'' -stars (see Lemma 3.7).

Note that, if y, y' are points on the rays $[o, \xi], [o, \xi']$, the CAT(0) comparison angle $\angle_{\bar{o}}(\overline{y}, \overline{y'})$ converges to the angular distance $d_T(\xi, \xi')$ as $y \rightarrow \xi$ and $y' \rightarrow \xi'$ (see e.g. [BH99, Proposition

II.9.8]). So, we may choose y, y' so that this CAT(0) comparison angle is at least $d_T(\xi, \xi') - \delta/8$. We have points $y_n \in [o, \xi_n)$ and $y'_n \in [o, \xi'_n)$ respectively converging to y, y' . See Figure 6.

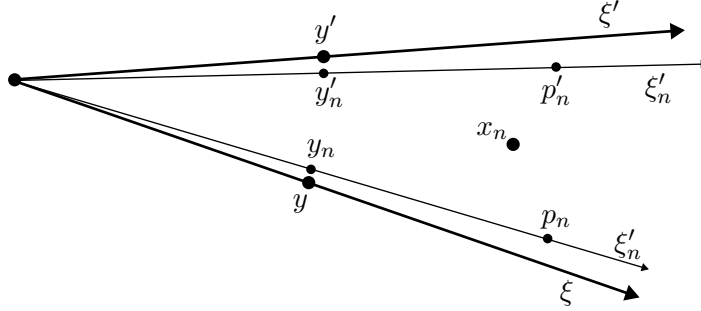


FIGURE 6. The rays $[o, \xi'_n), [o, \xi_n)$ converge to rays $[o, \xi), [o, \xi')$. By bounding $\angle_o(x_n, \xi_n)$ and $\angle_o(x_n, \xi'_n)$, we bound $\angle_o(\xi, \xi')$.

Since p_n and p'_n both leave every bounded subset of X , as long as n is large enough, y_n lies on the segment $[o, p_n]$ and y'_n lies on the segment $[o, p'_n]$, which implies that for large n the CAT(0) comparison angle $\angle_{\bar{o}}(\overline{p_n}, \overline{p'_n})$ satisfies

$$(5.4) \quad \angle_{\bar{o}}(\overline{p_n}, \overline{p'_n}) \geq d_T(\xi, \xi') - \delta/4.$$

We have chosen p'_n so that $d_X(p'_n, x_n) \leq D'$, which means that $d_X(o, p'_n) \geq Rn - B - D'$ and thus the comparison angle $\angle_{\bar{o}}(\overline{p'_n}, \overline{x_n})$ satisfies

$$\angle_{\bar{o}}(\overline{p'_n}, \overline{x_n}) \leq \sin^{-1} \left(\frac{D'}{Rn - B - D'} \right).$$

In particular for large enough n we have $\angle_{\bar{o}}(\overline{p'_n}, \overline{x_n}) \leq \delta/4$. We also observed above that $\angle_{\bar{o}}(\overline{p_n}, \overline{x_n}) \leq \delta$, yielding $\angle_{\bar{o}}(\overline{p_n}, \overline{p'_n}) \leq 5\delta/4$. Combining this with (5.4) we obtain $\angle(\xi, \xi') \leq 3\delta/2$.

As Θ'' was defined to be the δ -neighborhood of Θ' , our choice of δ ensures that $d_T(\Theta'', \sigma_{\text{mod}} - \text{ost}(\tau_{\text{mod}})) \geq 2\delta$. Since ξ belongs to $\text{st}_{\Theta''}(z)$ and ξ' belongs to $\text{st}(z')$, Lemma 3.8 implies that $z = z'$. $\square$

6. INTERPOLATION

We now complete the proof of the main theorem by applying Theorem 2.12 to the proto-coders constructed in the previous section. We have already done the work needed to verify that our proto-coders satisfy three out of the four conditions required by Theorem 2.12; the next proposition is the key to proving that the last condition also holds. In the statement below, when we say that a pair of sequences (x_n) and (y_m) in X are D -close infinitely often for some constant $D > 0$, we mean that there are infinitely many pairs of indices (m, n) so that $d_X(x_m, y_n) \leq D$.

Proposition 6.1 (Interpolation). *For any compact subset $\Theta \subset \text{ost}(\tau_{\text{mod}})$ and constants $K \geq 1, A \geq 0$, there is a compact set $\Theta' \subset \text{ost}(\tau_{\text{mod}})$ and constants $K' \geq 1, A', D \geq 0$ satisfying the following. For any pair of Θ -regular (K, A) -quasi-geodesic sequences $(x_n), (y_n)$ with the same τ_{mod} -limit $\tau \in M$, there is an “interpolating” Θ' -regular (K', A') -quasi-geodesic sequence (z_n) which is D -close infinitely often to both (x_n) and (y_n) .*

As a preliminary step, we prove the following.

Lemma 6.2. *Let τ be a τ_{mod} -simplex in $\partial_\infty X$, let Θ be a compact τ_{mod} -convex subset of $\text{ost}(\tau_{\text{mod}})$, and assume that Θ contains the barycenter of τ_{mod} . Then, for any subset $\Theta' \subset \text{ost}(\tau_{\text{mod}})$ containing a neighborhood of Θ , there exists a constant $C > 0$, depending on Θ' , such that for every $x, y \in X$,*

$$\{z \in V(y, \text{st}_\Theta(\tau)) : d_X(y, z) > Cd_X(x, y)\} \subset V(x, \text{st}_{\Theta'}(\tau)).$$

Proof. Fix $\delta > 0$ so that a δ -neighborhood of Θ is contained in Θ' , and let $\eta \in \partial_\infty X$ be the barycenter of τ . Let p be the point on $[y, \eta]$ where $[y, \eta]$ first enters $V(x, \text{st}_{\Theta'}(\tau))$. By Corollary 4.5,

$$d_X(y, p) < \left(\frac{1}{\sin \delta} + \frac{1}{\sin^2 \delta} \right) d_X(x, y).$$

Let $z \in V(y, \text{st}_\Theta(\tau))$. There exists $\xi \in \text{st}_\Theta(\tau)$ such that $z \in [y, \xi]$. Note that $V(y, \text{st}(\tau))$ is the union of sectors based at y containing the τ_{mod} -sector $V(y, \tau)$. In particular, there is a sector S containing both rays $[y, \eta]$ and $[y, \xi]$. Since $p \in [y, \eta]$, the ray $[p, \xi]$ is contained in S and is parallel to the ray $[y, \xi]$. On the other hand, p is contained in $V(x, \text{st}_{\Theta'}(\tau))$, so the ray $[p, \xi]$ is also contained in $V(x, \text{st}_{\Theta'}(\tau))$. By the convexity of the cone $V(x, \text{st}_{\Theta'}(\tau))$, any ray in the sector S starting from $p \in V(x, \text{st}_{\Theta'}(\tau))$ making an angle at most δ with the ray $[p, \xi]$ is contained in $V(x, \text{st}_{\Theta'}(\tau))$.

Let q be the last point along the geodesic $[y, \xi]$ such that $\angle_q(y, p) = \delta$. Note that if $z \in [y, \xi]$ satisfies $d_X(y, z) > d_X(y, q)$ then we have $\angle_z(p, q) \leq \delta$, hence $\angle_p(\xi, z) \leq \delta$, and therefore $z \in V(x, \text{st}_{\Theta'}(\tau))$. On the other hand, by the Euclidean law of sines,

$$d_X(y, q) = d_X(y, p) \frac{\sin \angle_p(y, q)}{\sin \delta} \leq \frac{d_X(y, p)}{\sin \delta},$$

and therefore

$$d_X(y, q) \leq \left(\frac{1}{\sin^2 \delta} + \frac{1}{\sin^3 \delta} \right) d_X(x, y).$$

Thus setting $C = \left(\frac{1}{\sin^2 \delta} + \frac{1}{\sin^3 \delta} \right)$ we see that if $d_X(x, z) > C d_X(x, y)$, we get $z \in V(x, \text{st}_{\Theta'}(\tau))$. $\square$

Proof of Proposition 6.1. By the Morse lemma (and Corollary 3.16), we may enlarge Θ, K, A slightly and replace (x_n) and (y_n) with uniformly close sequences satisfying $x_n \in V(x_0, \text{st}_\Theta(\tau))$ and $y_n \in V(y_0, \text{st}_\Theta(\tau))$ for all $n \geq 0$. By translating by an isometry of X and perturbing the sequences by another small amount, we can also assume that $x_0 = o$. We may also assume that Θ contains the barycenter of τ_{mod} .

Let Θ' be a τ_{mod} -convex compact subset of $\text{ost}(\tau_{\text{mod}})$ containing a neighborhood of Θ (so in particular Θ' contains a neighborhood of the barycenter of τ_{mod}). We shall inductively construct a Θ' -regular sequence (w_k) , whose terms alternate between (x_n) and (y_n) , with uniform gaps between consecutive terms. More precisely, we will find a sequence of indices (n_k) so that the sequence (w_k) defined by

$$w_k = \begin{cases} x_{n_k}, & k \text{ even} \\ y_{n_k}, & k \text{ odd} \end{cases}$$

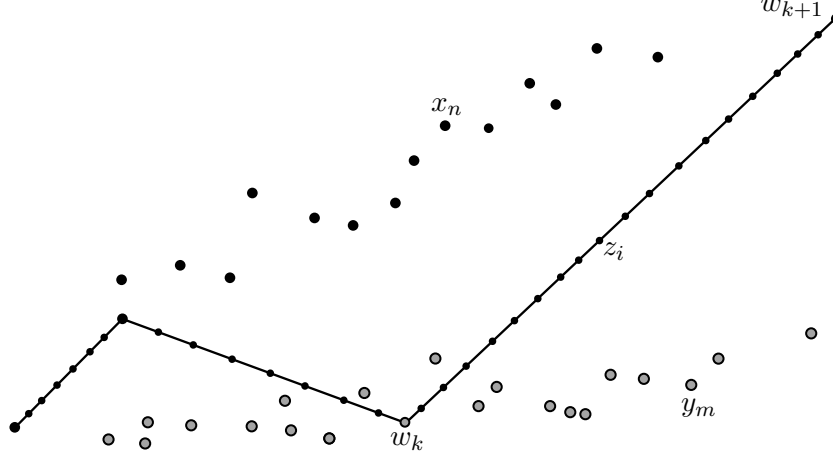
satisfies $w_{k+1} \in V(w_k, \text{st}_{\Theta'}(\tau))$ and $d_X(w_k, w_{k+1}) \geq 1$ for all k .

To construct (n_k) , set $n_0 = 0$, and suppose inductively that n_k (hence w_k) has already been defined for some $k \geq 0$; without loss of generality assume k is even, so $w_k = x_{n_k} \in V(x_0, \text{st}_{\Theta'}(\tau))$. Then define n_{k+1} so that $d_X(w_k, y_{n_{k+1}}) \geq 1$, and

$$n_{k+1} \geq \min\{m > n_k : y_m \in V(w_{n_k}, \text{st}_{\Theta'}(\tau)) \text{ for all } n \geq m\}.$$

Lemma 6.2 implies that the minimum above actually exists. Indeed, since Θ' contains a neighborhood of Θ and of the barycenter of τ_{mod} , there is a constant $C > 0$ so that $y_n \in V(w_{n_k}, \text{st}_{\Theta'}(\tau))$ whenever $d_X(y_n, y_0) > C d_X(w_{n_k}, y_0)$, and both distances $d_X(y_n, y_0)$ and $d_X(y_n, w_k)$ tend to infinity since (y_n) is a quasi-geodesic sequence.

We now construct the sequence (z_n) by “filling in the gaps” between consecutive terms in (w_k) using geodesic segments. That is, we let (z_n) be a sequence in X , starting with w_0 and containing (w_k) as a subsequence (z_{n_k}) , such that the terms in (z_n) appearing between $(z_{n_{k-1}})$ and (z_{n_k}) are evenly spaced along the geodesic $[z_{n_{k-1}}, z_{n_k}]$. Since $d_X(w_k, w_{k+1}) \geq 1$, we can do this while arranging that $1 \leq d_X(z_n, z_{n+1}) \leq 2$ for all n . See Figure 7.

FIGURE 7. The “interpolating” sequence (z_i) between the two sequences (x_n) and (y_m) .

We claim that the sequence (z_n) is Θ' -regular. To see this, first note that for any k , and any $n_{k-1} \leq n < m \leq n_k$, we have $z_m \in V(z_n, \text{st}_{\Theta'}(\tau))$ by convexity of Θ' -cones, hence $V(z_m, \text{st}_{\Theta'}(\tau)) \subset V(z_n, \text{st}_{\Theta'}(\tau))$ by the nesting property for Θ' -cones (Lemma 3.14). Now, for arbitrary indices $n < m$, fix indices $k \leq j$ such that $n_k \leq m < n_{k+1}$ and $n_j \leq m < n_{j+1}$. If $k = j$, then we get $V(z_m, \text{st}_{\Theta'}(\tau)) \subset V(z_n, \text{st}_{\Theta'}(\tau))$. Otherwise, we have $k < j$, hence $k + 1 \leq j$. By applying the nesting property for Θ' -cones inductively, we get

$$V(z_{n_j}, \text{st}_{\Theta'}(\tau)) = V(w_j, \text{st}_{\Theta'}(\tau)) \subseteq V(w_{k+1}, \text{st}_{\Theta'}(\tau)) = V(z_{n_{k+1}}, \text{st}_{\Theta'}(\tau)),$$

hence

$$V(z_m, \text{st}_{\Theta'}(\tau)) \subseteq V(z_{n_j}, \text{st}_{\Theta'}(\tau)) \subseteq V(z_{n_{k+1}}, \text{st}_{\Theta'}(\tau)) \subseteq V(z_n, \text{st}_{\Theta'}(\tau)).$$

This proves that $\theta(z_n \vec{z}_m) \in \Theta'$, so (z_n) is Θ' -regular.

It remains to prove that (z_n) is uniformly quasi-geodesic. The arguments above show that for any $n < m$, we have $z_m \in V(z_n, \text{st}_{\Theta'}(\tau))$, and thus we have a nested sequence of convex cones

$$V(z_0, \text{st}_{\Theta'}(\tau)) \supset V(z_1, \text{st}_{\Theta'}(\tau)) \supset \dots$$

It follows that the piecewise-geodesic path made up of the union of geodesic segments $[z_i, z_{i+1}]$ is Θ' -longitudinal in the sense of [KLP18, Def 3.10], so due to [KLP18, Lem. 3.11], it is uniformly bi-Lipschitz (with constant depending only on Θ') if it is parameterized by arc length. Since the distance between consecutive points in (z_n) is bounded between 1 and 2, this shows that (z_n) is a quasi-geodesic sequence. $\square$

Proof of Theorem 1.1. We construct a pair of proto-coders $\mathcal{S}, \mathcal{S}'$ satisfying the hypotheses of Theorem 2.12. Each proto-coder will come from a separate application of Proposition 5.1.

First we construct the proto-coder $\mathcal{S}$, which is chosen to satisfy condition 2 in Proposition 5.1. This proto-coder is adapted to the (standard) ρ_0 -action of Γ on M and generates a point coder $\mathcal{G}$. The proposition says that there is a compact subset $\Theta \subset \text{ost}(\tau_{\text{mod}})$ and constants $K \geq 1, A \geq 0$ so that if (g_n) is the path sequence for some $\mathcal{G}$ -coding of a flag $\tau \in M$, the sequence $(g_n o)$ is a Θ -regular (K, A) -quasi-geodesic, lying in a uniform neighborhood of $V(o, \text{st}(\tau))$. For this step, the first condition in Proposition 5.1 is irrelevant, so we can make an arbitrary choice of parameters in the hypotheses of the proposition.

Then, from Proposition 6.1, we can find a compact subset $\Theta' \subset \text{ost}(\tau_{\text{mod}})$ and constants $K' \geq 1, A' \geq 0$ so that if $(g_n)_{n=0}^\infty$ and $(h_m)_{m=0}^\infty$ are the path sequences associated to any two $\mathcal{G}$ -codings of the same point $\tau \in M$, there is a Θ' -regular (K', A') -quasi-geodesic sequence (x_ℓ) with τ_{mod} -limit

τ that is uniformly close to $(g_n o)$ infinitely often and uniformly close to $(h_m o)$ infinitely often. Using these parameters, we construct a new ρ_0 -adapted proto-coder $\mathcal{S}'$ satisfying condition 1 in Proposition 5.1. That is, if $\mathcal{G}'$ is the point coder generated by $\mathcal{S}'$, every Θ' -regular (K', A') -quasi-geodesic sequence in X has uniformly bounded Hausdorff distance from the orbit of a basepoint in X under the path sequence from some $\mathcal{G}'$ -coding.

We need to check that the proto-coders $\mathcal{S}, \mathcal{S}'$ satisfy all of the conditions (P1)-(P4) in Theorem 2.12. Conditions (P1), (P2), and (P3) follow immediately from Propositions 5.6, 5.8 and 5.9, respectively, so it remains to show that there is a finite set $F \subset \Gamma$ so that $\mathcal{S}$ and $\mathcal{S}'$ satisfy condition (P4). Let (g_n) and (h_m) be the path sequences associated to a pair of $\mathcal{G}$ -codings of the same point $\tau \in M$. As observed above, this implies that there is a Θ' -regular (K', A') -quasi-geodesic sequence (x_ℓ) with τ_{mod} -limit τ which is uniformly close to $(g_n o)$ infinitely often and uniformly close to $(h_m o)$ infinitely often.

Our construction of $\mathcal{S}'$ tells us that there is an $\mathcal{S}'$ -coding of a flag $\tau' \in M$, with path sequence $(f_k)_{k=0}^\infty$, so that the Hausdorff distance between $(f_k o)$ and (x_ℓ) is at most a constant $D > 0$ (independent of all of the codings). The second condition in Proposition 5.1 says that the τ_{mod} -limit of $(f_k o)$ is τ' , and Corollary 3.9 implies that (x_ℓ) also has τ_{mod} -limit τ' . Thus by uniqueness of τ_{mod} -limits we have $\tau = \tau'$.

Now, from the construction, there are infinitely many indices k so that $f_k o$ lies within uniform distance of some point of the form $g_n o$. Since Γ acts properly on X , we conclude that there is a fixed finite set F so that (f_k) and (g_n) are F -close. Arguing symmetrically, the same is true for (f_k) and (h_m) , and this concludes the proof of property (P4). $\square$

7. NECESSITY OF SEMI-CONJUGACY

In this section, we provide an example showing that in general it is not possible to replace the semi-conjugacies in Theorem 1.1 with actual (topological) conjugacies. For concreteness, we work in rank 2, although the construction can be generalized to any rank ≥ 2 . For a rank-one example, see [MMW22, Example 1.4].

7.1. The construction. Let $M = \mathbb{P}(\mathbb{Q}_p^3)$ be the p -adic projective plane, let $G = \text{PGL}_3(\mathbb{Q}_p)$, and let $\Gamma < G$ be a uniform lattice in G . In particular, Γ is finitely generated (hence countable).

For each point $z \in M$, let M_z denote the canonical blowup of M at z . The blowup M_z comes with a projection map $\pi_z : M_z \rightarrow M$, such that $\pi_z^{-1}(y)$ is a singleton for all $y \neq z$ and $\pi_z^{-1}(z)$ is identified with the projectivization of the tangent space to M at z (a copy of the p -adic projective line $\mathbb{P}(\mathbb{Q}_p^2)$). The blowup operation is natural in the sense that any $g \in G$ induces a homeomorphism $M_z \rightarrow M_{gz}$ (also denoted g) satisfying $\pi_{gz} \circ g = g \circ \pi_z$.

Now, fix a point $z \in M$ with trivial Γ -stabilizer, so the Γ -orbit $Z = \Gamma z$ is in bijection with Γ . To see that such a point exists, note that for each nontrivial element $g \in G$, the complement of the union of eigenspaces of g is an open dense subset of M , and then apply countability of Γ and the Baire category theorem.

Consider the countable family of maps $\mathcal{P} = \{M_z \rightarrow M : z \in Z\}$. This family can be organized into an (undirected) inverse system, and taking the inverse limit in the topological category yields a space $\hat{M}$ equipped with a projection $\hat{\pi} : \hat{M} \rightarrow M$. The Γ -action on Z defines a Γ -action of Γ on $\mathcal{P}$, and by the naturality of the projection maps π_z , this induces an action $\hat{\rho} : \Gamma \rightarrow \text{Homeo}(\hat{M})$. If we let $\rho_0 : \Gamma \rightarrow \text{Homeo}(M)$ denote the action on M coming from the inclusion of Γ into G , then $\hat{\rho}$ satisfies $\hat{\pi} \circ \hat{\rho}(\gamma) = \rho_0(\gamma) \circ \hat{\pi}$ for all $\gamma \in \Gamma$.

Observe that for each $z \in Z$, the blowup M_z is homeomorphic to a Cantor set. One way to see this is to let $L(z)$ denote the space of two-dimensional linear subspaces in $\mathbb{Q}_p^3$ containing z , and

realize M_z as the subspace

$$\{(w, \ell) \in M \times L(z) : w \subset \ell\}.$$

Thus M_z is identified with a nonempty perfect closed subspace of $M \times L(z)$. Since $M = \mathbb{P}(\mathbb{Q}_p^3)$ and $L(z) \simeq \mathbb{P}(\mathbb{Q}_p^2)$ are each homeomorphic to a Cantor set, it follows that M_z is nonempty, perfect, compact, metrizable, and totally disconnected, so it is also homeomorphic to a Cantor set.

This in turn implies that the limit $\hat{M}$ is also homeomorphic to a Cantor set, since $\hat{M}$ can be realized as the subspace

$$\left\{ (x_z) \in \prod_{z \in Z} M_z : \pi_z(x_z) = \pi_y(x_y) \quad \forall y, z \in Z \right\}.$$

This is a nonempty perfect closed subset of a countable product of Cantor sets, so it is homeomorphic to a Cantor set.

Now equip M with a metric d compatible with its topology. We have:

Proposition 7.1. *For every $\epsilon > 0$, there exists a homeomorphism $h : \hat{M} \rightarrow M$ satisfying*

$$d(h(x), \hat{\pi}(x)) < \epsilon,$$

for all $x \in \hat{M}$.

Proof. This is a consequence of the fact that, if $f : C \rightarrow C$ is any continuous surjective map of Cantor sets, there exists a homeomorphism $h : C \rightarrow C$ satisfying $d(h(x), f(x)) < \epsilon$ for any $x \in C$. To see that this holds, partition C into a finite collection of pairwise disjoint nonempty clopen subsets D_i , each with diameter at most ϵ . Each D_i is then homeomorphic to a Cantor set. Moreover, each preimage $f^{-1}(D_i)$ is a nonempty clopen subset of C , so it is also homeomorphic to a Cantor set, and we can define h on each $f^{-1}(D_i)$ to be an arbitrary homeomorphism to D_i . $\square$

For each $\epsilon > 0$, let $h_\epsilon : \hat{M} \rightarrow M$ be a homeomorphism as in the previous proposition. We can use h_ϵ to conjugate the action $\hat{\rho} : \Gamma \rightarrow \text{Homeo}(\hat{M})$ to an action $\rho_\epsilon : \Gamma \rightarrow \text{Homeo}(M)$.

Proposition 7.2. *For any fixed $\gamma \in \Gamma$, the homeomorphism $\rho_\epsilon(\gamma)$ converges to $\rho_0(\gamma)$ as $\epsilon \rightarrow 0$.*

Proof. Fix $x \in M$, and let $y = \hat{\pi}(h_\epsilon^{-1}(x))$. Then we have $d(x, y) = d(h_\epsilon(h_\epsilon^{-1}(x)), \hat{\pi}(h_\epsilon^{-1}(x))) < \epsilon$. By the definition of ρ_ϵ and $(\hat{\rho}, \rho_0)$ -equivariance of $\hat{\pi}$, we also have

$$d(\rho_\epsilon(\gamma)x, \rho_0(\gamma)y) = d((h_\epsilon \circ \hat{\rho}(\gamma) \circ h_\epsilon^{-1})(x), (\hat{\pi} \circ \hat{\rho}(\gamma) \circ h_\epsilon^{-1})(x)) < \epsilon.$$

Since $d(x, y) < \epsilon$, uniform continuity of $\rho_0(\gamma)$ implies that $d(\rho_0(\gamma)x, \rho_0(\gamma)y) \rightarrow 0$ uniformly as $\epsilon \rightarrow 0$. Thus $d(\rho_\epsilon(\gamma)x, \rho_0(\gamma)x) \leq d(\rho_\epsilon(\gamma)x, \rho_0(\gamma)y) + d(\rho_0(\gamma)y, \rho_0(\gamma)x)$ tends uniformly to zero. $\square$

Since Γ is finitely generated, this tells us that the ρ_ϵ -action lies in an arbitrarily small neighborhood of ρ_0 in $\text{Hom}(\Gamma, \text{Homeo}(M))$. Finally we show:

Proposition 7.3. *The action ρ_0 is not conjugate to ρ_ϵ for any $\epsilon > 0$.*

Proof. Since ρ_ϵ is conjugate to $\hat{\rho}$, it suffices to show that $\hat{\rho}$ is not conjugate to ρ_0 . We will show that the $\hat{\rho}$ -action is not cocompact on the set of pairs of distinct points in $\hat{M}$ (see Lemma 5.10).

First, define a metric $\hat{d}$ on $\hat{M}$ as follows. Fix an enumeration $\gamma_0, \gamma_1, \gamma_2, \dots$ of Γ , with $\gamma_0 = \text{id}$. Since z has trivial Γ -stabilizer, defining $z_n = \gamma_n z$ gives an enumeration of the orbit $Z = \Gamma z$. Fix an arbitrary metric d_0 inducing the correct topology on the blowup $M_z = M_{z_0}$, and for each n , let d_n be the metric on M_{z_n} obtained by pushing d_0 forward by the homeomorphism $\gamma_n : M_{z_0} \rightarrow M_{z_n}$. Then define $\hat{d}$ on the product $\prod_{n=0}^{\infty} M_{z_n}$ by

$$\hat{d}((x_n), (y_n)) = \sum_{n=0}^{\infty} 2^{-n} d_n(x_n, y_n).$$

The metric $\hat{d}$ induces the product topology. As we noted previously, $\hat{M}$ embeds into this product, so $\hat{d}$ restricts to a metric on $\hat{M}$ inducing the correct topology.

Now, for each $r > 0$, let $\hat{M}_r^{(2)}$ denote the set of pairs of distinct points in $\hat{M}$ whose $\hat{d}$ -distance is at most r . It suffices to prove that for every $r > 0$, there exists a pair of points whose Γ -orbit under the $\hat{\rho}$ -action is entirely contained in $\hat{M}_r^{(2)}$. Note that, with respect to the metric $\hat{d}$, the diameters of the fibers $\hat{\pi}^{-1}(z_n)$ tend to zero as $n \rightarrow \infty$. So, fix $N > 0$ so that, for all $n > N$, the diameter of the fiber $\hat{\pi}^{-1}(z_n)$ is less than r . Then, pick a pair of distinct points x, y in the fiber $\hat{\pi}^{-1}(z_0)$ so that for each element γ_n with $0 \leq n \leq N$, we have $\hat{d}(\hat{\rho}(\gamma_n)x, \hat{\rho}(\gamma_n)y) < r$. By construction, for every $\gamma \in \Gamma - \{\gamma_0, \dots, \gamma_N\}$, both points in the pair $\hat{\rho}(\gamma)x, \hat{\rho}(\gamma)y$ lie in some fiber $\hat{\pi}^{-1}(z_n)$ for $n > N$, and therefore also lie within distance r of each other. Thus the Γ -orbit of (x, y) lies in $\hat{M}_r^{(2)}$ and we are done. $\square$

Remark 7.4. The arguments in Proposition 7.1 and Proposition 7.2 above actually show the following general statement: if σ, ρ are two actions of the same finitely generated group on a Cantor set, and σ is semi-conjugate to ρ , then σ is conjugate to an arbitrarily small perturbation of ρ .

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